Journal of Computational Physics 556 (2026) 114814

Contents lists available at ScienceDirect

Journal of Computational Physics

journal homepage: www.elsevier.com/locate/jcp

A score-based diffusion model approach for adaptive learning of stochastic partial differential equation solutions

Phuoc Toan Huynh [a], Ruth Y. López Fajardo [a], Guannan Zhang [b], Lili Ju [c], Feng Bao [a,*]

[a] Department of Mathematics, Florida State University, Florida, Tallahassee, 32304, USA
[b] Computer Science and Mathematics Division, Oak Ridge Natioanl Laboratory, Oak Ridge, TN, 37831, USA
[c] Department of Mathematics, University of South Carolina, Columbia, SC, 29208, USA

ARTICLE INFO

Keywords:
Stochastic partial differential equation
Score-based diffusion model
Adaptive learning
Ensemble score filter
Data assimilation
Bayesian inference

ABSTRACT

We propose a novel framework for adaptively learning the time-evolving solutions of stochastic partial differential equations (SPDEs) using score-based diffusion models within a recursive Bayesian inference setting. SPDEs play a central role in modeling complex physical systems under uncertainty, but their numerical solutions often suffer from model errors and reduced accuracy due to incomplete physical knowledge and environmental variability. To address these challenges, we encode the governing physics into the score function of a diffusion model using simulation data and incorporate observational information via a likelihood-based correction in a reverse-time stochastic differential equation. This enables adaptive learning through iterative refinement of the solution as new data becomes available. To improve computational efficiency in high-dimensional settings, we introduce the ensemble score filter, a training-free approximation of the score function designed for real-time inference. Numerical experiments on benchmark SPDEs demonstrate the accuracy and robustness of the proposed method under sparse and noisy observations.

1. Introduction

In this paper, we introduce a score-based diffusion model approach for adaptively learning the time-evolving solutions of stochastic partial differential equations (SPDEs) through recursive Bayesian inference.

Partial differential equations (PDEs) are fundamental tools for modeling the dynamic behavior of complex physical systems. While they have been widely successful in scientific and engineering applications, many practical scenarios involve inherent uncertainties due to limited physical knowledge and environmental variability. For example, in climate and meteorological modeling, uncertainties in initial conditions, boundary data, and subgrid-scale physical processes can significantly affect the accuracy of predictions governed by PDEs such as the Navierâ€“Stokes or advectionâ€”diffusion equations. Similarly, in porous media flow problems, spatial heterogeneity and limited characterization of subsurface properties – such as permeability or porosity – introduce substantial uncertainty into models governed by Darcyâ€™s law and related PDEs, making accurate prediction particularly challenging. To capture these uncertainty effects and support reliable predictive analysis, it is essential to incorporate SPDEs into mathematical modeling framework. The numerical solution of SPDEs has thus become a central focus of the uncertainty quantification (UQ) community, where

* Corresponding author.
E-mail addresses: th24ba@fsu.edu (P.T. Huynh), rlopezfajardo@fsu.edu (R.Y. López Fajardo), zhangg@ornl.gov (G. Zhang), ju@math.usc.edu (L. Ju), fbao@fsu.edu (F. Bao).

https://doi.org/10.1016/j.jcp.2026.114814
Received 8 August 2025; Received in revised form 16 January 2026; Accepted 24 February 2026
Available online 1 March 2026
0021-9991/© 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

significant efforts have been dedicated to developing efficient solvers that can accurately characterize and propagate uncertainty in high-dimensional, nonlinear dynamical systems (see, e.g., [1–7] and the references therein).

Despite advances in SPDE solvers capable of quantifying uncertainty, significant challenges remain. In particular, model errors arising from limited physical knowledge and uncertainties due to unknown environmental variability can accumulate over time, leading to a progressive degradation in solution accuracy – especially when the underlying physical laws are incomplete or only approximately known [8–15].

A strategy to address these issues is to leverage measurement data to "learn" an estimated solution that reflects the underlying physical reality. However, fully data-driven approaches are often impractical due to the fact that real-world observations are typically sparse, indirect, and contaminated with noise. As a result, there is a critical need for adaptive learning frameworks that can effectively *integrate partial and noisy measurement data into existing physical models*. Such frameworks are essential for correcting model discrepancies, reducing uncertainty, and ultimately improving the fidelity of SPDE solvers.

The standard framework for integrating data and models to reduce the uncertainty and improve the estimation of the system state is known as "data assimilation", which systematically combines observational data with numerical model simulations to estimate the evolving state of a dynamical system. The mathematical formulation of this problem is referred to as "optimal filtering", which seeks to represent the "optimal" estimate of the target model state in the form of the *conditional expectation* – conditioning on the observational information. The state-of-the-art approach for solving the optimal filtering problem is the recursive Bayesian filter, including the Kalman type filters [10,16,17] and the particle filters [18–23]. In addition to the Bayesian approach, there exists a class of methods based on stochastic partial differential equations that directly approximate the conditional distribution of the target state [24–29].

In this work, we develop a novel framework that applies score-based diffusion models to the adaptive learning of SPDE solutions. Following the recursive Bayesian framework, our method used numerical SPDE solvers to encode first-principle and model-based dynamics for predictive simulation. When observational data becomes available, we apply Bayesian inference to integrate this data into the simulated model, enabling calibration of the model-based estimates. The adaptive learning in this context refers to the iterative refinement of the solution: as new data is assimilated, the model dynamically adjusts its estimates, progressively capturing hidden physics and reducing epistemic uncertainty. This leads to improved predictive accuracy and robustness over time, even in the presence of incomplete or imperfect knowledge of the underlying system.

The central contribution of this work is the introduction of a score-based diffusion model technique [30–32] that enables this adaptive learning process through data assimilation. While conventional optimal filtering methods have achieved considerable success, these methods face fundamental limitations. Kalman-type filters are efficient in high-dimensional settings but rely on linear or weakly nonlinear assumptions [10,33]. Particle filters, on the other hand, are well-suited for nonlinear systems but suffer from the "curse of dimensionality" issue in high-dimensional spaces due to its sequential Monte Carlo mechanism [34,35]. As a result, no existing method can effectively address the joint challenges of high dimensionality (e.g., arising from spatial discretization in SPDEs) and strong nonlinearity – both of which are intrinsic to adaptive learning tasks for SPDEs. Our framework is designed to overcome this gap by taking the advantage of the flexibility and scalability of score-based generative models [36].

In a score-based diffusion model, a forward stochastic differential equation (SDE) is used to progressively transform an arbitrary data distribution into a standard Gaussian distribution. During this process, the information from the original data is encoded in the so-called "score function", defined as the gradient of the log-probability density of the forward SDE. Then, a reverse-time SDE, driven by the score function, is solved to effectively transform samples drawn from the standard Gaussian distribution back into samples that follow the original data distribution. When applying score-based diffusion models to the adaptive learning of SPDE solutions, we encode the underlying physics described by the SPDE into the score function using simulation samples. When observational data becomes available, a likelihood score is introduced to incorporate data information into the reverse-time SDE, enabling the model to iteratively adjust its estimates and improve solution accuracy [37]. To enhance the computational efficiency of score-based diffusion models in high-dimensional settings and enable real-time calibration of estimated solutions, we propose an ensemble-based approximation of the score function-rather than relying on neural network-based deep learning methods [38]. We refer to this approach as the ensemble score filter.

We validate the proposed method through a series of numerical experiments involving prototypical SPDEs, including the Burgers' equations, incompressible Navierâ€“Stokes equations, and Allen-Cahn equations - all subjected to uncertainty perturbations. These tests demonstrate the accuracy, robustness, and adaptability of the proposed framework, particularly in scenarios with sparse and noisy observations. We emphasize that the EnSF formulation used in this work follows our earlier development in [37] and does not introduce new theoretical modifications. In [37], we primarily focused on establishing the score filter and EnSF methodology and illustrating it on a limited set of example problems. The main novelty of the present paper lies in the careful integration of EnSF with different types of numerical solvers for SPDEs to maintain stability and reliability in the data assimilation process, and in the systematic study of its performance on a range of classical PDE/SPDE benchmark problems.

The rest of the paper is organized as follows: Section 2 provides a detailed overview of the data assimilation framework for adaptively solving the SPDEs. We shall introduce the score filter methodology and its ensemble-based extension, namely the Ensemble Score Filter in Section 3. In Section 4, we present extensive numerical results to illustrate the effectiveness of the proposed approach. Some concluding remarks will be given in Section 5.

2. Adaptive learning for stochastic partial differential equations via data assimilation

2.1. Partial differential equations with uncertainty

We consider the following partial differential equations:

$$du_t(x) = \mathcal{L}(x,t)dt + h(u_t, x, t)dt, \tag{1}$$

where u is the unknown function (either scalar or vector-valued) defined over the spatial domain $x \in \Omega \subset \mathbb{R}^{dim}$, $u_t := u(\cdot, t)$ and $t \geq 0$ is the time variable. The operator $\mathcal{L}$ is a second-order (typically elliptic) spatial differential operator, and $h(u_t, x, t)$ represents a given source term.

In practical applications, various uncertainties arise due to limited physical knowledge and incomplete environmental information. As a result, the deterministic PDE (1) is often subject to random perturbations, leading to the following stochastic partial differential equation (SPDE):

$$du_t(x) = \mathcal{L}(\tilde{\omega}, x, t)dt + h(u_t, \tilde{\omega}, x, t)dt + \sigma_t d\tilde{W}(x,t), \tag{2}$$

where $\tilde{\omega}$ encapsulates *model uncertainties due to incomplete environmental information*, and we formulate this kind of uncertainties by unknown coefficients in the spatial differential operator and the source term. The term $\tilde{W}(x,t)$ denotes a space-time Wiener process that captures *model errors arising from unresolved or unknown physical processes* during the formulation of the PDE model, and σ_t is the noise coefficient.

Extensive research in the uncertainty quantification (UQ) community has focused on developing efficient numerical algorithms for characterizing and propagating the uncertainties inherent in stochastic partial differential equations (SPDEs) (see, e.g., [1–7,27] and the reference therein). To simplify our presentation, we use the following generic form to represent the numerical solver that propagates the approximate solution of the SPDE (2):

$$u_{t_{n+1}} = \mathbb{F}(u_{t_n}, t_n, \tilde{\omega}, \Delta\tilde{W}_{t_n}), \tag{3}$$

where $\Delta\tilde{W}_{t_n} := \tilde{W}_{t_{n+1}} - \tilde{W}_{t_n}$, and $\mathbb{F}$ denotes a forward operator that incorporates the random variables $\tilde{\omega}$ and $\Delta\tilde{W}_{t_n}$ within a UQ-enabled PDE solver. This operator advances the solution u in Eq. (2) from time t_n to t_{n+1}, with $u_{t_{n+1}}$ being a numerical approximation to $u(\cdot, t_{n+1})$. In the numerical experiments section (Section 4), we provide a more detailed discussion of the various forward solvers used for SPDE solution propagation, depending on the specific types of PDEs being solved.

While most conventional UQ techniques aim to characterize the uncertainties inherent in SPDE systems [39–45], the model errors and uncertainties stemming from limited knowledge and incomplete information remain unavoidable. *The primary goal of this work is to introduce a robust and accurate data assimilation framework based on score-based diffusion models, designed to adaptively reduce model uncertainties and mitigate system errors by leveraging partial and noisy observations of the target system.*

In what follows, we briefly outline the data assimilation framework and explain how it can be applied to adaptively enhance the accuracy of SPDE solutions.

2.2. A data assimilation framework for adaptive learning of SPDE solutions

The data assimilation problem combines a spatial-temporal model typically derived from a forward numerical solver with an observational system. Specifically, we consider the following formulation:

$$\begin{aligned} X_{t_{n+1}} &= f(X_{t_n}, \omega_n), && \text{(State)} \\ Y_{t_{n+1}} &= g(Y_{t_n}) + \xi_n. && \text{(Measurement)} \end{aligned} \tag{4}$$

Here, $X_{t_n} := [u_{t_n}(x^1), u_{t_n}(x^2), \ldots, u_{t_n}(x^d)]$ denotes the discretized state vector of the quantity of interest. In this work, the state corresponds to the solution of the SPDE, evaluated at a set of spatial grid points $\{x^i\}_{i=1}^d$ over the domain $\Omega \in \mathbb{R}^{dim}$.

In the state equation, the function f represents the forward model, which incorporates the random variable ω_n to account for uncertainties. When this forward model is implemented through a numerical solver for the SPDE (3), the random input includes both model uncertainty $\tilde{\omega}$ and model error $\tilde{W}_{t_n}$.

As the forward model propagates in time using UQ-enabled SPDE solvers, uncertainties and errors naturally accumulate over time, leading to progressively less accurate predictions with reduced confidence. In practice, these uncertainties often arises from stochastic components with unknown distributions, making precise solution prediction infeasible. To mitigate this degradation in accuracy, observational data – typically collected via measurement instruments – are incorporated to calibrate the model and enhance solver accuracy. However, such observations are generally sparse, noisy, and incomplete, and therefore insufficient by themselves to fully reconstruct the underlying state.

The measurement equation in the data assimilation framework directly reflects the need for observational data to calibrate and correct the forward model predictions. As described in Eq. (4), the observed data Y are related to the true target state X through an observation operator g, with additive measurement noise ξ_n. These observations help constrain the uncertainty in the model and improve the accuracy of state estimates. Throughout this work, we assume that the observational noise ξ_n is Gaussian with covariance matrix R.

The goal of data assimilation is to recover the best possible estimate of the target state given the available observational data $Y_{t_1:t_{n+1}} := \left\{Y_{t_1}, \ldots, Y_{t_{n+1}}\right\}$. In the context of adaptive learning for SPDEs considered in this work, data assimilation plays a critical

role in reducing uncertainties arising from incomplete information and model errors. The mathematical framework that solves the data assimilation problem is the *optimal filter*, which seeks the "best estimate" in the form of the conditional expectation of $X_{t_{n+1}}$ given $Y_{t_1:t_{n+1}}$. Specifically, the optimal filter $\tilde{X}_{t_{n+1}}$ is defined as

$$\tilde{X}_{t_{n+1}} := \mathbb{E}\Big[X_{t_{n+1}} | Y_{t_1:t_{n+1}}\Big].$$

In practical implementations, rather than computing the conditional expectation directly, one typically aims to derive an approximation of the conditional distribution $P(X_{t_{n+1}}|Y_{t_1:t_{n+1}})$, referred to as the *"filtering distribution"*.

The standard mathematical framework for computing the filtering distribution is recursive Bayesian filter. Suppose that at time step n, the filtering distribution $P(X_{t_n}|Y_{1:t_n})$ is available, representing the current estimate of the state given all observations up to time t_n. The Bayesian filter then proceeds in two steps: a prediction step and an update step.

In the prediction step, we utilize the Chapman-Kolmogorov formula to obtain

$$P(X_{t_{n+1}}|Y_{t_1:t_n}) = \int P(X_{t_{n+1}}|X_{t_n})P(X_{t_n}|Y_{t_1:t_n})dX_{t_n}, \qquad \textit{Prior filtering distribution} \tag{5}$$

where $P(X_{t_{n+1}}|X_{t_n})$ is the transition probability derived from the state dynamics (4) governed by the forward solver f. The prior filtering distribution $P(X_{t_{n+1}}|Y_{t_1:t_n})$ contains the data information up to time t_n and the model information captured by the transition probability.

In the update step, we apply the following Bayesian inference scheme to compute the posterior filtering distribution of the state variable:

$$P(X_{t_{n+1}} \mid Y_{t_1:t_{n+1}}) \propto P(X_{t_{n+1}} \mid Y_{t_1:t_n})P(Y_{t_{n+1}} \mid X_{t_{n+1}}), \qquad \textit{Posterior filtering distribution} \tag{6}$$

which allows us to adaptively refine the estimate of the SPDE solution. The likelihood function $P(Y_{t_{n+1}}|X_{t_{n+1}})$ quantifies the probability of the observed data given the current state:

$$P(Y_{t_{n+1}}|X_{t_{n+1}}) \propto \exp\Big[-\frac{1}{2}(g(X_{t_{n+1}}) - Y_{t_{n+1}})^\top R^{-1}(g(X_{t_{n+1}}) - Y_{t_{n+1}})\Big], \tag{7}$$

where R is the covariance matrix associated with the observational noise introduced in the measurement process described in Eq. (4).

Together, the prediction and update steps form a recursive model-data integration framework that enables adaptive learning of the system state – specifically, the solution of the SPDE – over time. In the prediction stage, the filtering distribution is propagated forward using the transition probability $P(X_{t_{n+1}}|X_{t_n})$, which encodes the model dynamics and uncertainty. However, without incorporating observational data, model errors and uncertainty would accumulate, leading to increasingly inaccurate predictions. The update step plays a critical role in adaptively correcting these predictions by assimilating new observations through Bayesian inference. At the same time, the model information embedded in the prior filtering distribution helps reconstruct the unobserved components of the solution that are not captured by the indirect and potentially sparse measurement process. This recursive framework allows the system to adaptively refine its state estimates by incorporating new data, dynamically adjusting both the solution and its associated uncertainty in response to discrepancies between model predictions and observations.

In the following section, we shall introduce an accurate and robust ensemble score filter approach for solving the data assimilation problem.

3. The ensemble score filter approach for data assimilation

The score-based filter methodology characterizes the filtering distribution in data assimilation problems using the score function within the diffusion model framework. To begin, we briefly outline the design of the score filter for data assimilation in Section 3.1. We then introduce an ensemble-based Monte Carlo approximation scheme for the score function and derive the resulting ensemble score filter method in Section 3.2.

3.1. Score filter for data assimilation

In the score-based diffusion model framework, two stochastic processes are defined as follows:

$$\begin{aligned} dZ_\tau &= b_\tau Z_\tau d\tau + \sigma_\tau dW_\tau, & Z_0 &= Q_0(X), & \text{(Forward SDE)} \\ dZ_\tau &= \big(b_\tau Z_\tau - \sigma_\tau^2 S(Z_\tau,\tau)\big)d\tau + \sigma_\tau d\overline{W}_\tau, & Z_T &= N(0, \boldsymbol{I}_d). & \text{(Reverse SDE)} \end{aligned} \tag{8}$$

The first equation is a forward stochastic differential equation (SDE) that propagates a given target data distribution $Z_0 = Q_0(X)$, such as the filtering distribution in data assimilation, toward the standard Gaussian distribution over the pseudo time interval $[0, 1]$. The second equation is the corresponding time-reversed SDE, which transports the standard Gaussian random variable $Z_1 = N(0, \boldsymbol{I}_d)$ back to the original data distribution Z_0. The drift coefficient b and diffusion coefficient σ in the forward SDE are pre-defined deterministic functions. In this work, we let $b_\tau = \frac{d\log\alpha_\tau}{d\tau}$ and $\sigma_\tau = \sqrt{\frac{d\beta_\tau^2}{d\tau} - 2\frac{d\log\alpha_\tau}{d\tau}\beta_\tau^2}$ with $\alpha_\tau = 1 - \tau$ and $\beta_\tau = \tau$ (see [37]).

The key component that enables the reverse SDE in Eq. (8) to transport samples from the standard Gaussian distribution to the target data distribution is the *score function* $S(Z_\tau, \tau)$, defined by

$$S(Z_\tau, \tau) = \nabla_z \log Q_\tau(Z_\tau), \tag{9}$$

where Q_τ is the distribution of the forward SDE solution Z_τ. In other words, the score function S acts as a guide that steers the reverse-time sampling process in the diffusion model framework back toward the original data distribution.

The methodology of the score filter is to connect the target filtering distribution to a score function and generate state samples that follow the filtering distribution through the reverse SDE process.

Specifically, suppose that we have access to the posterior score $S_{n|n}$ associated with the posterior filtering distribution $P(X_{t_n} \mid Y_{t_1:t_n})$. By simulating the reverse SDE in Eq. (8) using this score, we can generate samples that follow the filtering distribution. Let $\{x_{t_n}^{(j)}\}_{j=1}^J$ denote the set of generated samples, where J is any positive integer. To incorporate model dynamics, we propagate these state samples forward using the state evolution model in Eq. (4) and get

$$x_{t_{n+1}}^{(j)} = f(x_{t_n}^{(j)}, \omega_n^{(j)}), \qquad j = 1, 2, \dots, J.$$

Recall that in the context of adaptive learning for SPDEs, the state evolution function f represents the numerical forward solver of the SPDE under consideration.

The state samples $\{x_{t_{n+1}}^{(j)}\}_{j=1}^J$ carry information from both the previous posterior filtering distribution and the state dynamics (i.e, the PDE model). These samples are used to approximate the prior filtering score, denoted by $S_{n+1|n}$, which plays a central role in the prediction step of the recursive Bayesian filtering procedure. This score is then used to generate samples from the prior filtering distribution $P(X_{t_{n+1}} \mid Y_{t_1:t_n})$. The approximation of $S_{n+1|n}$ can be accomplished using deep learning techniques [38]. However, a more computationally efficient approach in data assimilation is to use a Monte Carloâ€“based ensemble approximation [31,37], which will be detailed in the following subsection.

Once the prior score $S_{n+1|n}$ is available, we incorporate the observational data to update the filtering distribution. To this end, we define the posterior score $S_{n+1|n+1}(Z_\tau, \tau)$ through the following scheme:

$$S_{n+1|n+1}(Z_\tau, \tau) := S_{n+1|n}(Z_\tau, \tau) + h(\tau) S_{LH}(Z_\tau, \tau), \qquad \tau \in [0, 1], \tag{10}$$

where $S_{LH}(Z_\tau, \tau) := \nabla_z \log P(Y_{t_{n+1}} \mid X_{t_{n+1}})$ is referred to as the *"likelihood score"*. This term represents the gradient of the log-likelihood function, which quantifies the discrepancy between the model prediction and the observation, and has the form of a score function. The function $h(\cdot)$ is a damping function satisfying $h(0) = 1$ and $h(1) = 0$.

Accordingly, the terminal condition for Eq. (10) at $\tau = 0$ is given by

$$S_{n+1|n+1}(Z_0, 0) = S_{n+1|n}(Z_0, 0) + \nabla_z \log P(Y_{t_{n+1}} \mid X_{t_{n+1}}),$$

which corresponds to the Bayesian update expressed in Eq. (6). In contrast, the terminal condition at $\tau = 1$ is

$$S_{n+1|n+1}(Z_1, 1) = S_{n+1|n}(Z_1, 1),$$

reflecting the fact that $Z_1 \sim N(0, \boldsymbol{I}_d)$. This completes the update step of the filtering procedure.

Note that the choice of the damping function $h(\tau)$ governs how the observational data (encoded in the likelihood score) is integrated with the model dynamics (encoded in the prior score). In this work, we adopt the simple linear form $h(\tau) = 1 - \tau$, for $\tau \in [0, 1]$. More advanced choices for $h(\tau)$ will be explored in future studies.

The remaining challenge in the score-based filtering framework lies in the approximation of the (prior) score function itself, which we now address in the following subsection.

3.2. Monte Carlo-base ensemble approximation for the score function

The methodology of the ensemble score filter (EnSF) builds on the fact that the distribution of the forward SDE solution can be expressed as follows [38]:

$$Q_\tau(Z_\tau) = \int_{\mathbb{R}^d} Q_\tau(Z_\tau \mid Z_0) Q_0(Z_0) dZ_0,$$

where the transition probability $Q_\tau(Z_\tau \mid Z_0)$ over the pseudo-temporal interval $\tau \in [0, 1]$ is given by

$$Q_\tau(Z_\tau \mid Z_0) = N(\alpha_\tau Z_0, \beta_\tau^2 \boldsymbol{I}_d)$$

due to the linearity of the diffusion process described in Eq. (8).

With the above expressions, the score function defined in Eq. (9) can be rewritten as follows [38,46]:

$$\begin{aligned} S(Z_\tau, \tau) =& \nabla_z \log \left(\int_{\mathbb{R}^d} Q_\tau(Z_\tau \mid Z_0) Q_0(Z_0) dZ_0 \right) \\ =& \int_{\mathbb{R}^d} -\frac{Z_\tau - \alpha_\tau Z_0}{\beta_\tau^2} w(Z_\tau, Z_0) Q_0(Z_0) dZ_0, \end{aligned} \tag{11}$$

where the weight function $w(Z_\tau, Z_0)$ is defined by

$$w(Z_\tau, Z_0) := \frac{Q_\tau(Z_\tau \mid Z_0)}{\int_{\mathbb{R}^d} Q_\tau(Z_\tau \mid \tilde{Z}_0) Q_0(\tilde{Z}_0) d\tilde{Z}_0},$$

and satisfies the normalization condition $\int_{\mathbb{R}^d} w(Z_\tau, Z_0) Q_0(Z_0) dZ_0 = 1$.

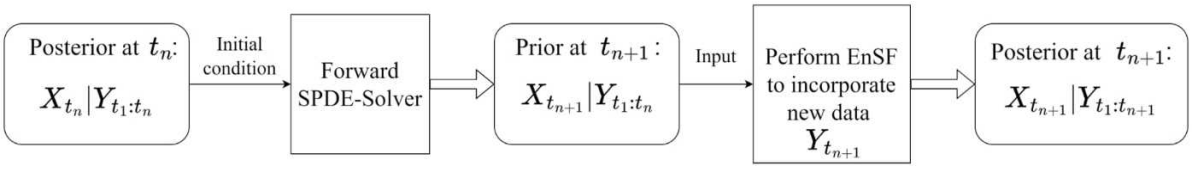


Fig. 1. Workflow of the EnSF applying to SPDEs.

The integral representation of the score function described in Eq. (11) naturally leads to a Monte Carloâ€“based ensemble approximation:

$$S(Z_\tau, \tau) \approx \sum_{j=1}^{J} \frac{Z_\tau - \bar{\alpha}_\tau z_0^{(j)}}{\bar{\beta}_\tau^2} \iota\bar{w}(Z_\tau, z_0^{(j)}), \tag{12}$$

where $\{z_0^{(j)}\}_{j=1}^J$ are samples that serve as "data samples" in the diffusion model to characterize the target distribution Q_0 [47]. The modified diffusion parameters are defined as $\bar{\alpha}_\tau := 1 - \tau(1-\epsilon_\alpha)$ and $\bar{\beta}_\tau := \sqrt{\epsilon\beta + \tau(1-\epsilon_\beta)}$, where ϵ_α and ϵ_β are regularization hyper-parameters introduced to stabilize the reverse SDE dynamics.

In the context of data assimilation for adaptive learning of SPDE solutions, the data samples are used to represent the predicted state variable through simulating the SPDE model via the UQ-enabled PDE solver f. Specifically, the prior score $S_{n+1|n}$ associated with the prior filtering distribution $P(X_{t_{n+1}} \mid Y_{t_1:t_n})$ can be approximated as

$$S_{n+1|n}(Z_\tau, \tau) \approx \textstyle\sum_{j=1}^{J} \frac{Z_\tau - \bar{\alpha}_\tau \bar{x}_{t_{n+1}}^{(j)}}{\bar{\beta}_\tau^2} \iota\bar{w}(Z_\tau, \bar{x}_{t_{n+1}}^{(j)}), \tag{13}$$

where each sample $\bar{x}_{t_{n+1}}^{(j)}$ is generated via $\bar{x}_{t_{n+1}}^{(j)} = f(x_{t_n}^{(j)}, \omega_n^{(j)})$, $j = 1, 2, \ldots, J$. The approximated prior score is then used to compute the posterior score according to the formulation in Eq. (10). We then solve the reverse-time SDE in (8) with the newly computed posterior score to generate the sample $\left\{x_{t_{n+1}}^{(j)}\right\}_{j=1}^J$ that approximate the posterior filtering distribution $P\left(X_{t_{n+1}}|Y_{t_1:t_{n+1}}\right)$. This sample is used to give an estimation of the SPDE solution at time instance t_{n+1}. We conclude this section by providing the workflow of our proposed algorithm in Fig. 1 and showing the pseudo-code of the algorithm in Algorithm 1.

In the next section, we present detailed numerical implementations of the EnSF method to demonstrate its adaptive learning mechanism across a range of SPDE models.

Algorithm 1 EnSF algorithm for SPDEs.

Input: the numerical solver $\mathbb{F}$ that represents the forward solver f, the observation function $g(X)$, distribution $P(X_{t_0})$ as the guess for the initial state of the SPDE;
Generate J samples $\left\{x_{t_0}^{(j)}\right\}_{j=1}^J$ from the prior $P(X_{t_0})$;
for $n = 0, \ldots,$ **do**
    Generate J samples $\left\{\bar{x}_{t_{n+1}}^{(j)}\right\}_{j=1}^J$ via $\bar{x}_{t_{n+1}}^{(j)} = f(x_{t_n}^{(j)}, \omega_n^{(j)})$, $j = 1, 2, \ldots, J$ that approximate $P\left(X_{t_{n+1}}|Y_{t_1:t_n}\right)$;
    Perform the EnSF:
        Compute $\bar{S}_{n+1|n}$ using (13) with $\left\{\bar{x}_{t_{n+1}}^{(j)}\right\}_{j=1}^J$;
        Compute $\bar{S}_{n+1|n+1}$ using (10);
        Solve the reverse SDE in (8) to generate J samples $\left\{\tilde{x}_{t_{n+1}}^{(j)}\right\}_{j=1}^J$ that approximate $P\left(X_{t_{n+1}}|Y_{t_1:t_{n+1}}\right)$;
    Set $\hat{x}_{t_{n+1}} = \frac{1}{J}\sum_{j=1}^{J} \tilde{x}_{t_{n+1}}^{(j)}$ to be the estimation at time t_{n+1} and generate the samples $x_{t_{n+1}}^{(j)} = \tilde{x}_{t_{n+1}}^{(j)}$ for $j = 1, \ldots, J$ for the next filtering step.
end for

4. Numerical experiments

We conduct numerical experiments to evaluate the effectiveness of our proposed adaptive learning approach in improving the accuracy of SPDE solvers across three classical test problems. In addition to demonstrating the performance of our method, these examples also highlight its flexibility to be combined with different types of numerical methods. Moreover, our framework is implemented as a versatile package that allows interested researchers to plug in their own solvers to achieve modelâ€“data integration. In the first example, we study the two-dimensional (2D) Burgers' equations and examine the performance of our method in capturing sharp features that develop in the solution. The second example involves 2D Navierâ€–Stokes equations with various boundary conditions and external forces. We also account for different levels of model error introduced in the forward solver. Finally, in the third example we study the 2D Allenâ€–Cahn equations with general mobility, exploring multiple ways to incorporate model uncertainties.

For all cases, we simplify our demonstration by consider (3) in the form

$$u_{t_{n+1}} = \mathbb{F}(u_{t_n}, t_n, \tilde{\omega}) + \sigma_n \tilde{W}_{t_n}, \tag{14}$$

where $\mathbb{F}$ denotes the forward operator that propagates the solution of the SPDE in time, $\tilde{W}_{t_n} \sim N(0, \boldsymbol{I}_d)$ is a standard Gaussian random variable, representing a generic model uncertainty, and the noise magnitude is scaled by the coefficient σ_n. The random variable $\tilde{\omega}$ captures various uncertainties due to unknown physics or simulation errors. In all experiments, the state vector u_{t_n} is high-dimensional, with dimension on the order of 10^3 for Taylor-Green vortex problem and 10^4 for the other test cases, so our data assimilation procedure operates in a high-dimensional state space. We also consider a highly nonlinear observation operator of arctangent type, as well as scenarios with extremely sparse spatial observations. The performance of our method in these challenging settings is compared against the local ensemble transform Kalman filter (LETKF), a state-of-the-art approach for data assimilation. These experiments are therefore designed precisely to test our method in regimes with high dimensionality and strong nonlinearity, as well as sparse and noisy observations, which we identify as the main focus and contribution of this work. We shall present a brief explanation of the implementation of the operator $\mathbb{F}$ for each numerical experiment.

4.1. Burgers' equation

Next, we consider the following 2D Burgers' equation:

$$\begin{aligned} &u_t + \left(\frac{u^2}{2}\right)_x + \left(\frac{u^2}{2}\right)_y = 0, \quad (x, y, t) \in (-1, 1)^2 \times (0, T), \\ &u_0(x, y) = \tfrac{1}{4} + \tfrac{1}{4}\sin(\pi(x + y)), \quad (x, y) \in (-1, 1)^2, \end{aligned} \tag{15}$$

subject to periodic boundary conditions. It is well-known for this type of equations that even though the initial condition is smooth, the solution can develop sharp corners over time and eventually becomes discontinuous (see, e.g., [48]). Therefore, it is important to choose an appropriate numerical solver to achieve reliability and stability in the reference solution as well as in the data assimilation procedure. There exist several high-resolution schemes that can handle discontinuities in solutions of the Burgers' equation. In this work, we construct the forward solver that propagates the PDE model by employing a Runge-Kutta discontinuous Galerkin (RKDG) method to solve (15) numerically. Specifically, (15) is discretized in space by a discontinuous Galerkin (DG) method [49] and in time using strong-stability-preserving Runge-Kutta (SSP-RK) schemes [50]. To handle moving shocks of the numerical solution after a finite time, the numerical solver is combined with the standard TVB (total variation bounded) slope limiter [51]. **We emphasize that the TVB slope limiter is also essential for the data assimilation process since without it, noise accumulation can destabilize the simulation and lead to numerical blow-up.**

This RKDG-TVB combination is a widely used and well-studied approach for Burgers-type equations, supported by well-established theoretical results. Our adaptive learning and data assimilation framework, however, is not tied to this specific solver. One can choose other methods that are compatible with the DG framework and the TVB slope limiter, such as the fully explicit localized exponential time-differencing Runge-Kutta (ETDRK) schemes developed in [52], or other discretization techniques such as finite-volume (weighted) essentially non-oscillatory (ENO/WENO) schemes [48,53–59]. However, ETDRK schemes require the computation of matrix exponentials, and ENO/WENO schemes demand a large number of stencil points due to polynomial reconstructions. These features become expensive in the data assimilation setting, where we must generate a large ensemble of forward predictions, which motivates our choice of the RKDG-TVB solver in this study.

In this numerical experiment, we aim to demonstrate the performance of our method for two terminal times: $T = 0.2$ when the solution is smooth and $T = 0.45$ when the solution starts to exhibit sharp corners. Moreover, since the mass of the solution to (15) is preserved over time (see, e.g., [52]), we also focus on investigating whether the estimated solutions achieve similar behavior.

Regarding the discretization of the model problem, we keep the mesh size fixed at $h = 1/40$ for spatial discretization in the numerical solvers, while using different time-step sizes for the reference solution and for data assimilation. However, as the SSP-RK is an explicit time-stepping scheme, those time steps must be chosen carefully to satisfy the CFL conditions. Specifically, our synthetic "true state" is obtained by solving the fully discrete version of (15) on a very fine temporal mesh with mesh size $\Delta t = T/800$, while the filtering process is performed on a coarser temporal mesh with $\Delta t_{Filter} = T/80$. For simplicity, we only consider the second-order DG and SSP-RK methods. The discretized version of (15) is also used in the forecast model. To ensure the stability of our estimates, we apply the TVB scheme to the analysis at each filtering step. More details of the implementation can be found on Github https://github.com/Toanhuynh997/StateEst_PDEs/tree/main/Burgers2D.

For this Burgers' equation example, the uncertainty in the model is incorporated by adding the term $\sigma_n \tilde{W}_{t_n}$ to the numerical solver $\mathbb{F}$, with $\sigma_n = 0.01$. Additionally, uncertainty in the initial condition is introduced by sampling the initial state as $u_0 \sim 2 \cdot N(0, \mathbf{I_d})$. We point out that at each filtering time t_n, we add the perturbation term $\sigma_n \tilde{W}_{t_n}$ to the current forecast and then feed this perturbed state into the solver to advance to the next time step. Consequently, each realization evolves along its own stochastic trajectory rather than the deterministic path of the reference solution. Thus, in our data assimilation procedure we solve an SPDE instead of the deterministic PDE used to generate the reference.

Observations are collected under three scenarios: (i) full observation of the state (100%), (ii) partial observation covering 60% of the state, and (iii) **sparse observation with only** 10% **of the state is observed**. In all cases, the observational data of the reference (true) solutions are obtained through the ***arctangent operator***, i.e.,

$$Y_t = \arctan(X_t) + \varepsilon_t, \quad t = 1, 2, \ldots, N, \quad \text{with} \quad \varepsilon_t \sim \mathcal{N}(\vec{0}, 0.1^2 I_d).$$

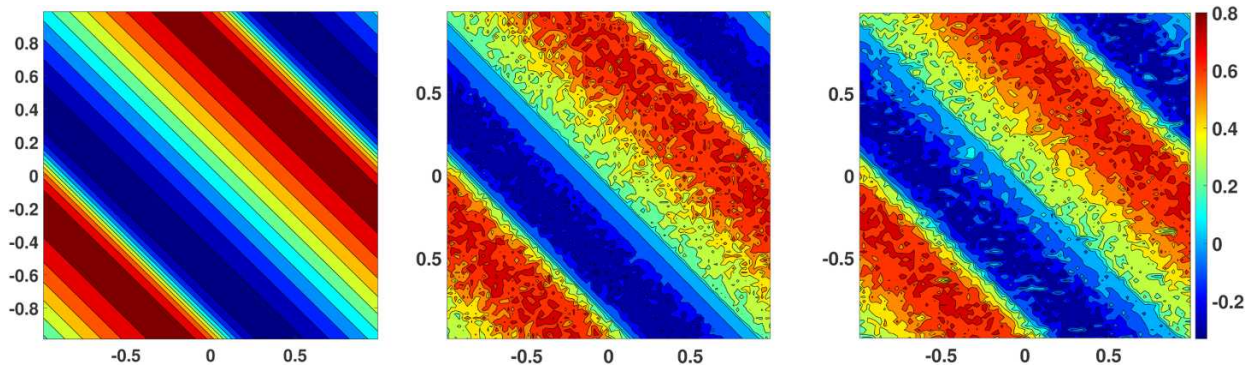


Fig. 2. Contour maps of the estimated solution to the Burgers' equation at $T = 0.2$. (Left) Reference solution. (Center) EnSF estimate with 100% observations. (Right) EnSF estimate with 60% observations.

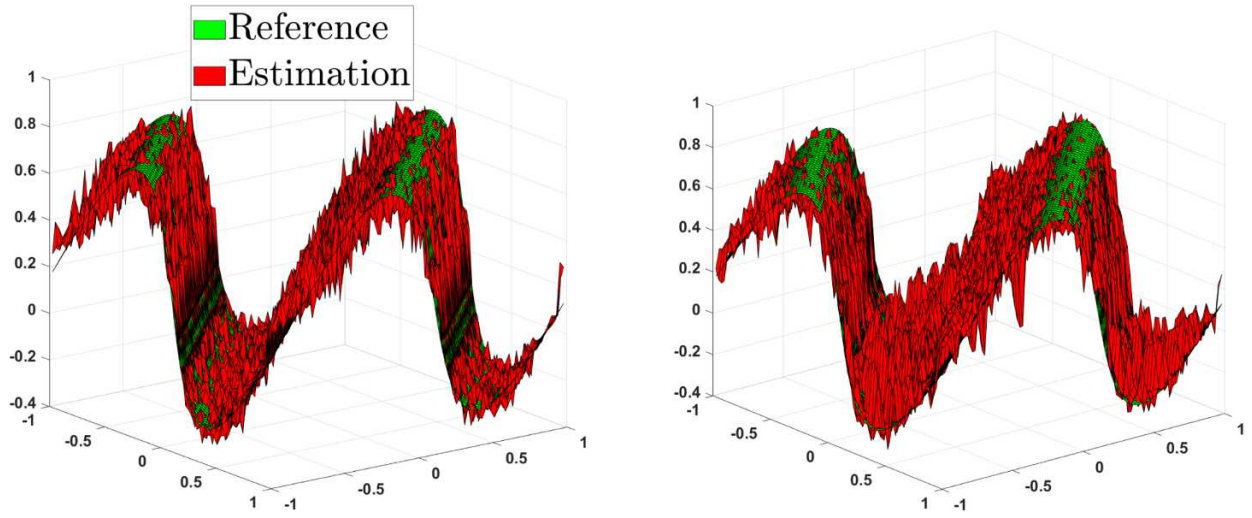


Fig. 3. 3D view of the estimated solution for the Burgers' equation at $T = 0.2$. (Left) Estimate with 100% observations. (Right) Estimate with 60% observations.

Note that the arctangent observation operator is highly nonlinear and is used here as a simplified proxy to reflect key challenges present in real-world measurement processes, such as those arising in radar or satellite observations. For the 10% observation case, we also compare the performance of our method with the LETKF. Regarding the setting of reversed SDE in the EnSF, the number of ensembles and the time steps are chosen to be 80 and 300 for the cases with 100% and 60% observations, while for the case with 10%, they are designated to be 100 and 400, respectively.

We first evaluate the performance of the EnSF in cases where the solution of the Burgers' equation is fully observed (100%) and partially observed (60%). The contour plots of the EnSF-estimated solutions at $T = 0.2$ and $T = 0.45$ are presented in Figs. 2 and 4, respectively. To provide a cleared view of the performance of the solution estimation, in Figs. 3 and 5 we display the 3D visualizations of the reference (true) solution alongside the corresponding EnSF estimates.

Figs. 2–5 demonstrate that the EnSF is capable of providing highly accurate solution estimates. When only 60% of the solution is observed, the estimation becomes less accurate due to the reduced amount of observational data.

Finally, the evolution of the corresponding masses and the Root Mean Square Errors (RMSEs) for solution estimation over time are displayed in Fig. 6 (for a total time of $T = 0.2$) and Fig. 7 (for $T = 0.45$). Although the initial mass does not match the reference solution due to discrepancies in the guessed initial condition, the mass conservation property is quickly recovered. As for the RMSEs, both cases show a quick decrease over time. However, minor fluctuations appear in the RMSE curves after $t = 0.3$ in the $T = 0.45$ case, which is expected, as this is when sharp corners begin to develop in the solution.

To make the problem more challenging, we consider a scenario in which only 10% of the reference solution is observed. In this case, we apply the EnSF-based data assimilation method to estimate the solution under uncertainty and sparse observational coverage. To address the issue of sparse observation, we incorporate a PDE-based inpainting technique into the EnSF framework. Specifically, we employ the biharmonic (Bih) inpainting method to reconstruct unobserved regions of the state [31].

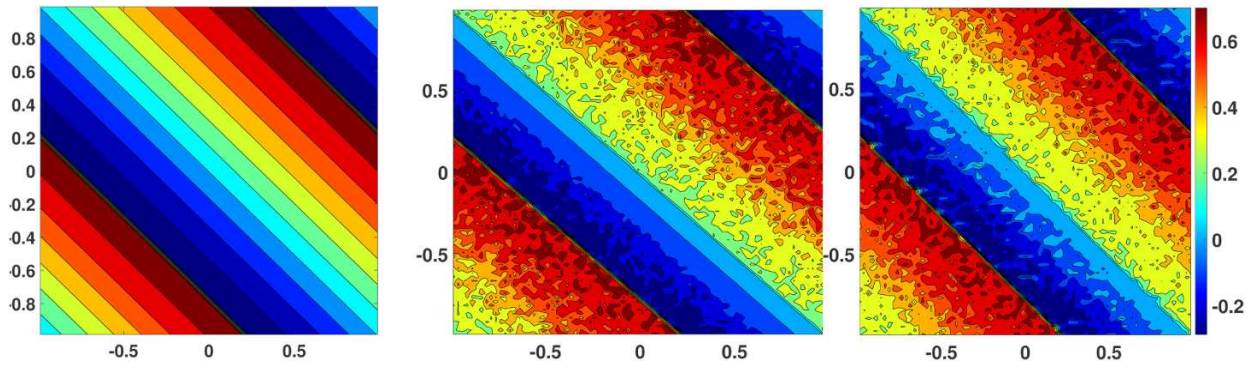


Fig. 4. Contour maps of the estimated solution to the Burgers' equation at $T = 0.45$. (Left) Reference solution. (Center) EnSF estimate with 100% observations. (Right) EnSF estimate with 60% observations.

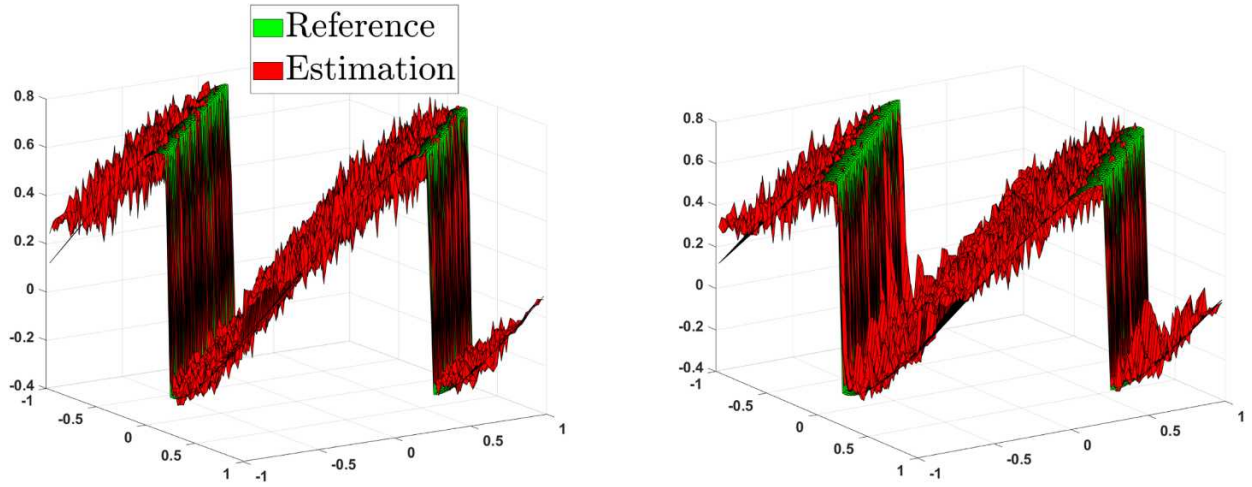


Fig. 5. 3D view of the estimated solution for the Burgers' equation at $T = 0.45$. (Left) Estimate with 100% observations. (Right) Estimate with 60% observations.

We compare the estimation performance obtained from two approaches: by using EnSF with the Bih inpainting and by LETKF. The estimated solutions for $T = 0.2$ and $T = 0.45$ are shown in Figs. 8 and 11, respectively. The three-dimensional views are depicted in Figs. 9 and 12. The corresponding mass evolution and RMSEs are displayed in Figs. 10 and 13, respectively. The displayed figures show that the EnSF approach yields more accurate solution estimates and outperforms the LETKF. Moreover, by incorporating the inpainting technique, the estimated solutions appear significantly less noisy, even though only 10% of the spatial domain was observed.

The above results show that the proposed framework reduces forecast error and mitigates model uncertainty in the predicted states. Nevertheless, because EnSF produces an ensemble of posterior estimates conditioned on noisy and sparse observations, a nonzero level of variability remains across ensemble members. To quantify this residual variability during filtering, we report the mean and variance (or standard deviation) of the RMSE over the ensemble for the 10% observation setting at final times $T = 0.2$ and $T = 0.45$. These RMSE curves are displayed in Fig. 14.

4.2. Navier-Stokes equation

In this subsection, we focus on the incompressible Navier-Stokes (NS) equations given as follows:

$$\begin{cases} \boldsymbol{u}_t - \nu\Delta\boldsymbol{u} + \boldsymbol{F}(\boldsymbol{u}) + \nabla p = \boldsymbol{f}, & \text{in } \Omega \times (0, T], \\ \nabla \cdot \boldsymbol{u} = 0, & \text{in } \Omega \times (0, T], \end{cases} \tag{16}$$

subject to the initial condition $\boldsymbol{u}(\boldsymbol{x}, 0) = \boldsymbol{u}_0(\boldsymbol{x})$ and the Dirichlet or the periodic boundary conditions. Ω in (16) is an open bounded domain in $\mathbb{R}^2$, $\boldsymbol{u} = (u, v)$ and p are the unknown velocity field and pressure, respectively, $\boldsymbol{F}(\boldsymbol{u}) = (\boldsymbol{u} \cdot \Delta)\boldsymbol{u}$ denotes the nonlinear convection, $\boldsymbol{f}$ is an external force, and ν represents the kinematic viscosity which is inversely proportional to the Reynolds number Re. Similar to the Burgers' equation, we aim to choose a numerical method that is efficient to implement and produce numerical solutions that satisfy the desired physical property, which in this test case is energy dissipation. Several existing schemes meet these requirements, such as the scalar auxiliary variable (SAV) schemes [60–65] and the Lagrange multiplier schemes [66,67]. However, there

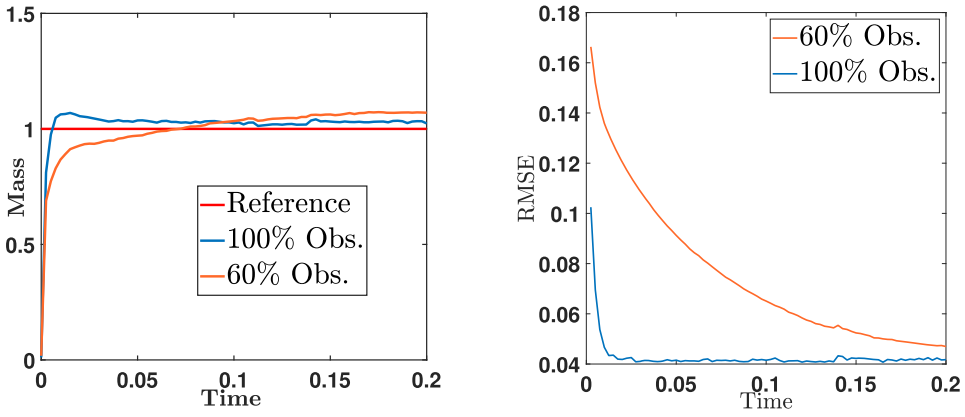


Fig. 6. (Left) Evolution of mass with $T = 0.2$. (Right) RMSE for state estimations with $T = 0.2$.

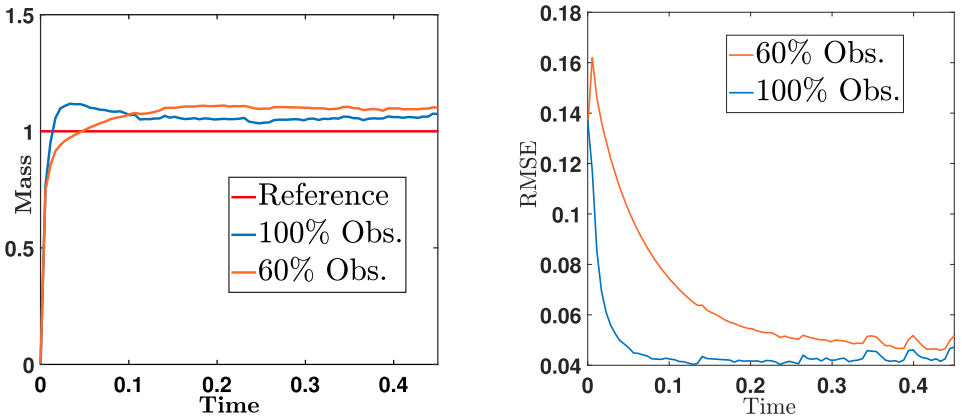


Fig. 7. (Left) Evolution of mass with $T = 0.45$. (Right) RMSE for state estimations with $T = 0.45$.

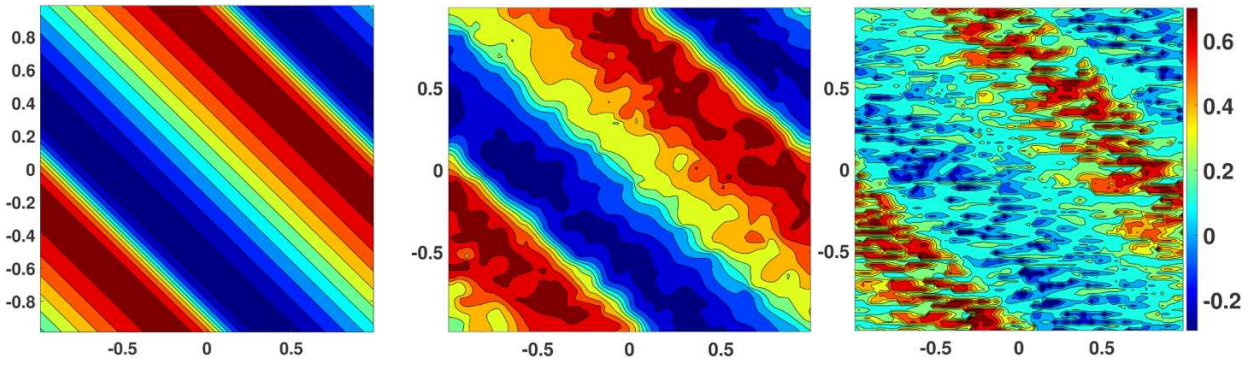


Fig. 8. Contour maps of the estimated solution to the Burgers' equation at $T = 0.2$ with 10% observations. (First) Reference solution. (Second) EnSF estimate with biharmonic inpainting. (Third) LETKF estimate.

remain challenges that may hinder the data assimilation process. For example, both approaches require sufficiently small time-step sizes to maintain accuracy, and the Lagrange multiplier approach demands solving nonlinear algebraic equations for the Lagrange multiplier at each time step, which may produce multiple real or even complex solutions [68]. Therefore, to formulate the forward solver, we employ a newly developed dynamically regularized Lagrange multiplier (DRLM) method proposed in [68] to discretize

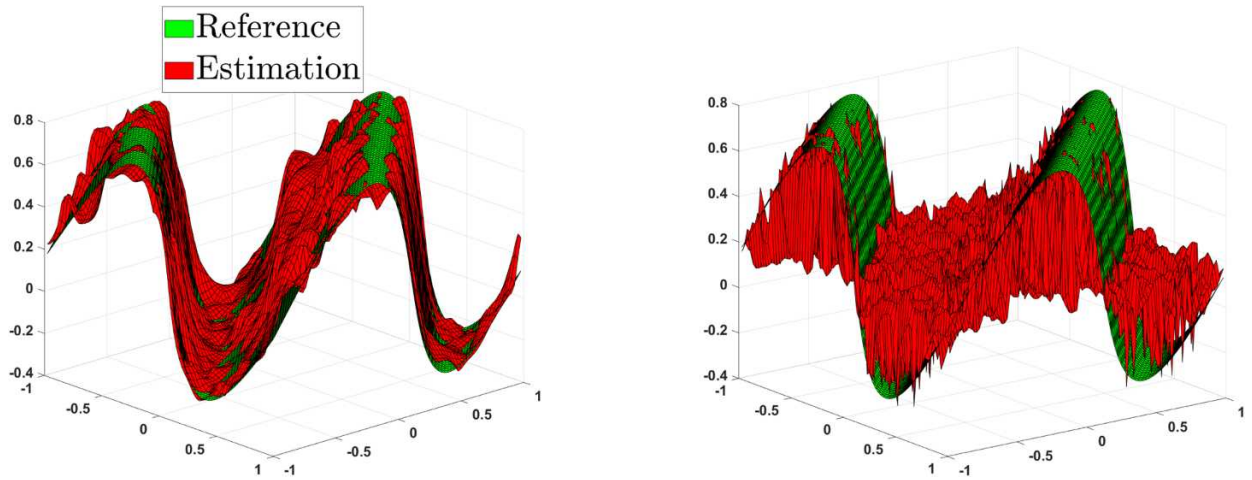


Fig. 9. 3D view of the estimated solution for the Burgers' equation at $T = 0.2$ with 10% observations. (Left) EnSF estimate with biharmonic inpainting. (Right) LETKF estimate.

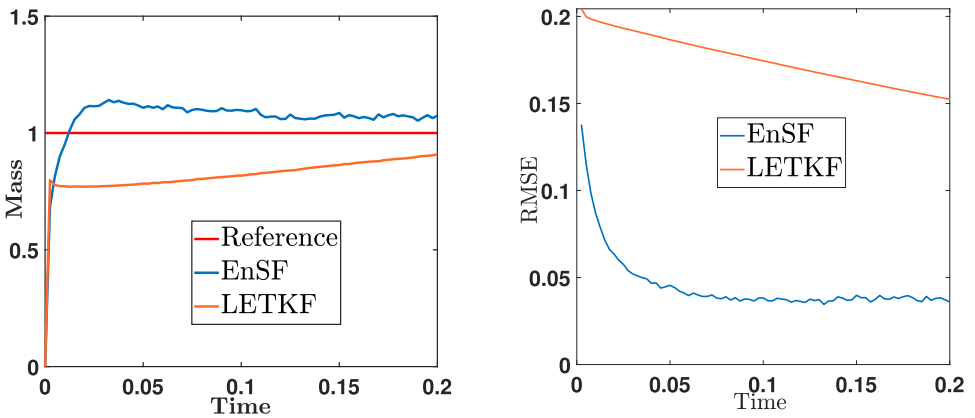


Fig. 10. (Left) Evolution of Mass with $T = 0.2$ with 10% observations. (Right) RMSE for state estimations with $T = 0.2$ with 10% observations.

the equations (16). More specifically, an artificial Lagrange multiplier q is introduced to reformulate (16) into

$$\begin{cases} \boldsymbol{u}_t - \nu\Delta\boldsymbol{u} + \boldsymbol{F}(\boldsymbol{u}) + \nabla p = \boldsymbol{f}, & \text{in } \Omega\times(0,T], \\ \nabla\cdot\boldsymbol{u} = 0, & \text{in } \Omega\times(0,T], \\ \frac{d\boldsymbol{\mathcal{K}}(\boldsymbol{u})}{dt} + \theta\frac{dq^2}{dt} = \big(\boldsymbol{u}_t + q\boldsymbol{F}(\boldsymbol{u}),\boldsymbol{u}\big), & \text{in } \Omega\times(0,T], \end{cases} \tag{17}$$

where $\boldsymbol{\mathcal{K}}(\boldsymbol{u}) = \frac{1}{2}\int_\Omega |\boldsymbol{u}|^2 d\boldsymbol{x}$ is the kinetic energy associated with (16), $q(0) = 1$ and $\theta > 0$ is a regularization parameter. It is straightforward to verify that at the continuous level, we have $q(t) = 1$ for all $t > 0$, and that the new system (17) is equivalent to the original one [68]. We discretize the reformulated Navier-Stokes system (17) in space using a finite-difference scheme on a MAC staggered grid [69] and in time with backward-differentiation-formula (BDF) methods. Alternatively, one may substitute the finite-difference spatial discretization with a finite-element or finite-volume approach and adopt the Crank-Nicolson scheme for time stepping. We emphasize that the DRLM schemes employed in this work require solving two generalized Stokes systems at each time step. One can instead replace the Stokes solver with two successive Poisson solvers via a pressure-correction approach [67,70]. For a complete derivation of the fully discrete formulations under both strategies, see [68]. Accordingly, we augment the filter state with a scalar variable $\tilde{q}$ to represent the estimate of q. Since q is an artificial parameter with known true value $q = 1$, we simply use that true value as the observation for $\tilde{q}$.

Throughout the experiments presented in this example, different time-discretization schemes are used for generating the "true state" and for prediction during the filtering process. Specifically, the synthetic reference solution representing the "true state" is computed using the second-order DRLM solver, while the first-order DRLM scheme is used during prediction to reduce computational cost. The detailed numerical implementation can be found on https://github.com/Toanhuynh997/StateEst_PDEs/tree/main/NavierStokes .

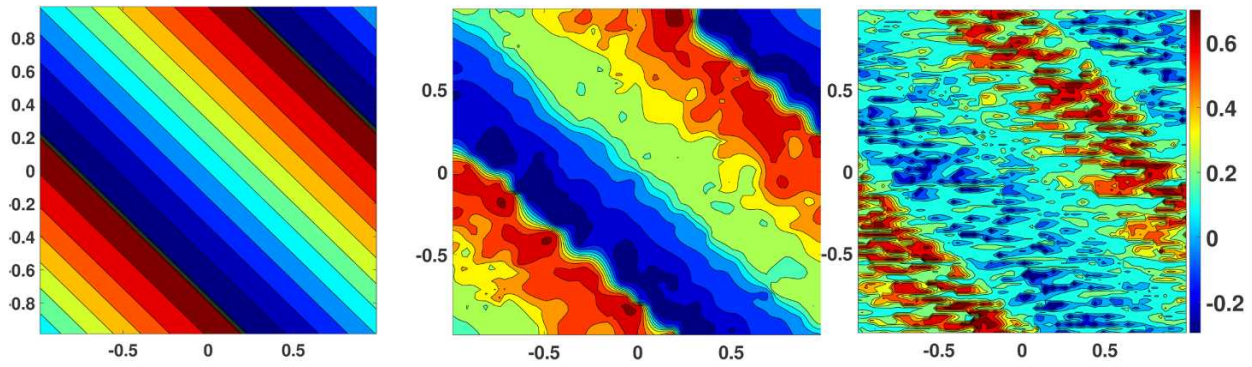


Fig. 11. Contour maps of the estimated solution to the Burgers' equation at $T = 0.45$ with 10% observations. (First) Reference Solution. (Second) EnSF estimate with bi-harmonic inpainting. (Third) LETKF estimate.

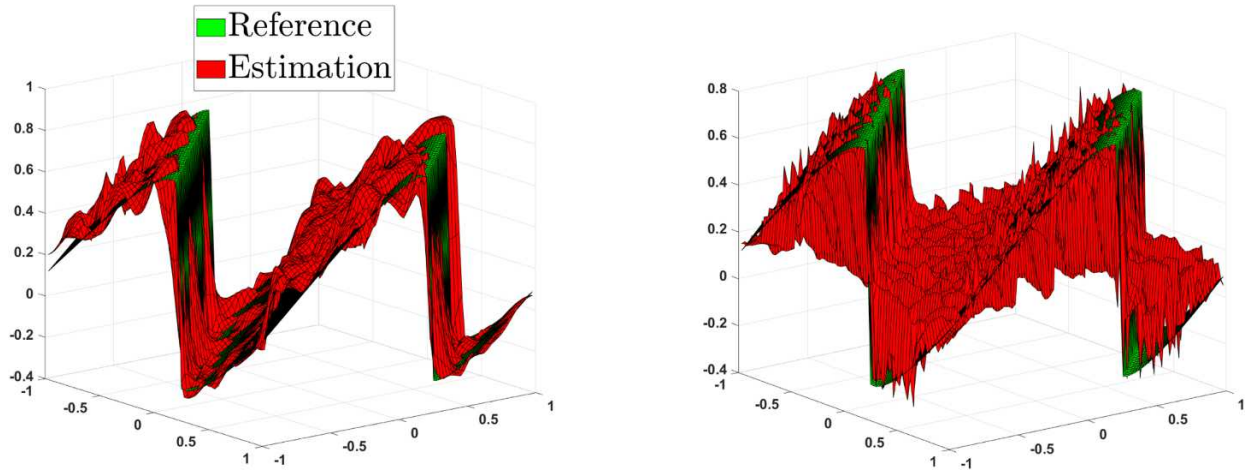


Fig. 12. 3D view of the estimated solution for the Burgers' equation at $T = 0.45$ with 10% observations. (Left) EnSF estimate with biharmonic inpainting. (Right) LETKF estimate..

4.2.1. *Taylor-Green vortex problem*

We begin with the Taylor-Green vortex problem on the domain $\Omega = (0, 1)^2$ where Eq. (16) is considered with zero external forcing, i.e., $\boldsymbol{f} = [0, 0]^T$, and periodic boundary conditions are imposed. The Reynolds number is set at $Re = 1000$, the parameter θ is fixed at $\theta = 5$, and the final time is $T = 1$, and the mesh size is fixed at $h = 1/40$. To generate the reference solution as the "true state", we solve Eq. (16) numerically using the second-order DLMR method on a fine temporal grid with time step $\Delta t = T/600$. Observations are then obtained by applying the ***arctangent operator*** to this reference solution.

In this experiment, we let the initial conditions for the reference solution be

$$\begin{cases} u(x, y) = \sin(2\pi x)\cos(2\pi y), \\ v(x, y) = -\cos(2\pi x)\sin(2\pi y). \end{cases}$$

The reference pressure field and the reference velocity field are illustrated in Fig. 15. To implement the prediction step in data assimilation, we solve the PDE on a coarser temporal grid with a time step of $\Delta t_{Filter} = T/100$. The initial conditions for u and v are assumed to follow a Gaussian distribution: $2 \cdot N(0, \boldsymbol{I}_d)$. We explore two levels of additive model uncertainties by setting $\tilde{W}^i_{t_n} \sim \sigma^i_n N(0, \boldsymbol{I}_d), i = 1, 2$ where $\sigma^1_n = 0.001$ and $\sigma^2_n = 0.1$. Similarly to the Burgers' equation, the perturbed term $\sigma_n \tilde{W}^i_{t_n}$ is also added to the predicted solution at each filtering step, which results in solving an SPDE and forces the forecast to evolve in a different trajectory compared to the reference one.

It is well known that, in the absence of the external forces, the solution to Eq. (16) under the prescribed boundary conditions satisfies the energy dissipation [67,68]. Therefore, in addition to state estimation, we also investigate whether the estimated solution fulfills the discrete energy dissipation law.

To proceed, **we consider three levels of observational coverage:** 100%, 70%, **and** 7%. As in the previous example, the estimated solutions produced by our method for the 7% observation case will be compared with those obtained using the LETKF.

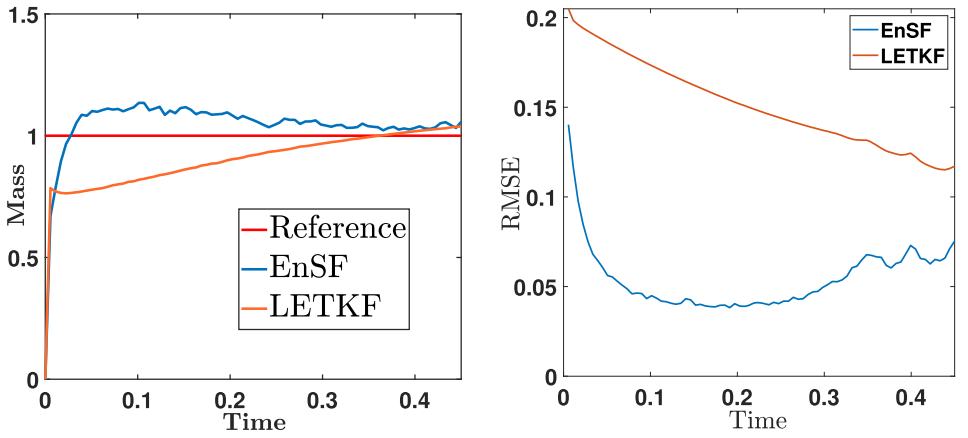


Fig. 13. (Left) Evolution of Mass with $T = 0.45$ with 10% observations. (Right) RMSE for state estimations with $T = 0.45$ with 10% observations.

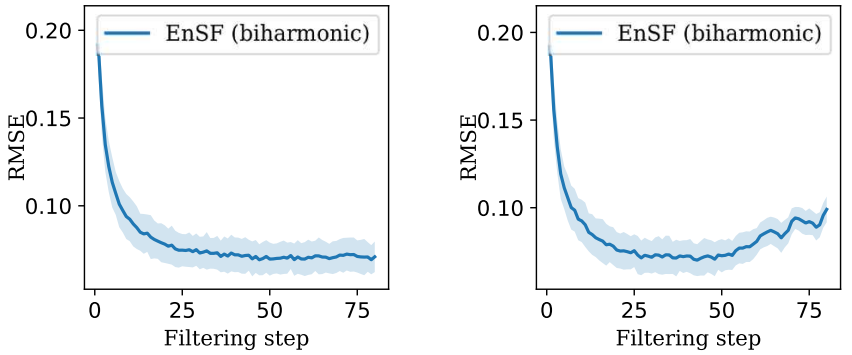


Fig. 14. Mean and variance of the RMSE (across ensemble members) for Burgersâ€™equation with 10% observations. (Left) $T = 0.2$. (Right) $T = 0.45$.

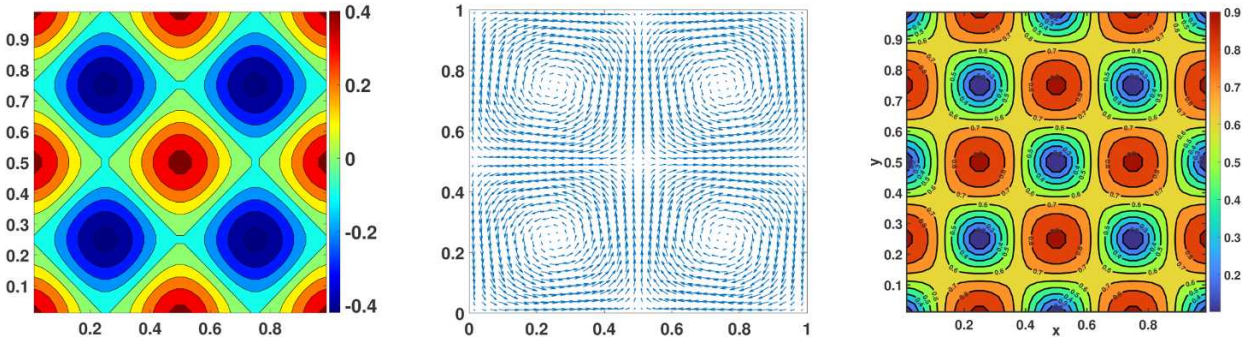


Fig. 15. Reference solution for the Taylor-Green vortex problem. (Left) Pressure field. (Center) Velocity field. (Right) Velocity magnitude.

We begin with presenting the results with 100% and 70% observations. The estimated solutions with 100% amount of observations are displayed in Figs. 16 and 17, while those obtained with 70% are illustrated in Figs. 18 and 19. The corresponding estimated energies are shown in Fig. 20.

It can be observed that our proposed method recovers accurately the behavior of the reference solution across two types of noise. Moreover, as shown in Fig. 20, although the estimated energy initially deviates from the reference, it quickly converges and begins to decrease after roughly 10 filtering iterations for the 100% observation case and about 20 iterations for the 70% observation case.

We conclude the Taylor-Green vortex test case by presenting the results with only 7% observations. Due to the sparsity of observations, the Bi-harmonic inpainting technique was also applied to the EnSF in this experiment. The results corresponding to the noise $\tilde{W}_{t_n}^1$ are displayed in Figs. 21–23. Specifically, Fig. 21 illustrates the performance of the EnSF in state estimation, Fig. 22 shows the performance of the LETKF in state estimation, and Fig. 23 presents comparisons of the estimated energies and the RMSEs for state estimation. Similarly, results for the noise level $\tilde{W}_{t_n}^2$ are shown in Figs. 24–26, following the same structure: EnSF estimates in Fig. 24, LETKF estimates in Fig. 25, and energy/RMSE comparisons in Fig. 26.

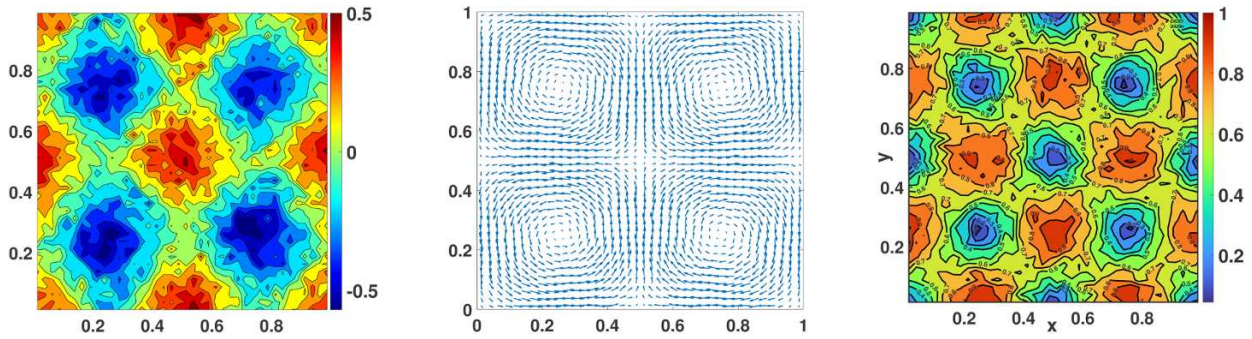


Fig. 16. Estimated state for the Taylor-Green vortex problem with 100% observations and model uncertainty level σ_n^1. (Left) Pressure field. (Center) Velocity field. (Right) Velocity magnitude.

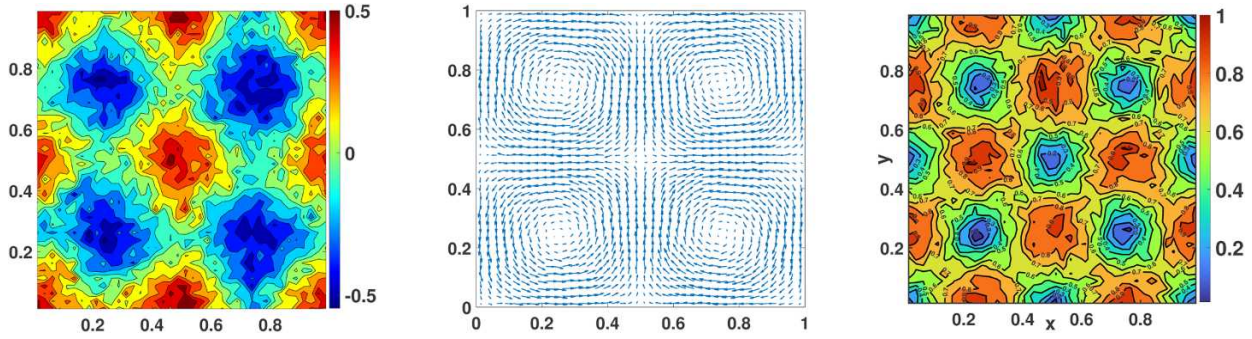


Fig. 17. Estimated state for the Taylor-Green vortex problem with 100% observations and model uncertainty level σ_n^2. (Left) Pressure field. (Center) Velocity field. (Right) Velocity magnitude.

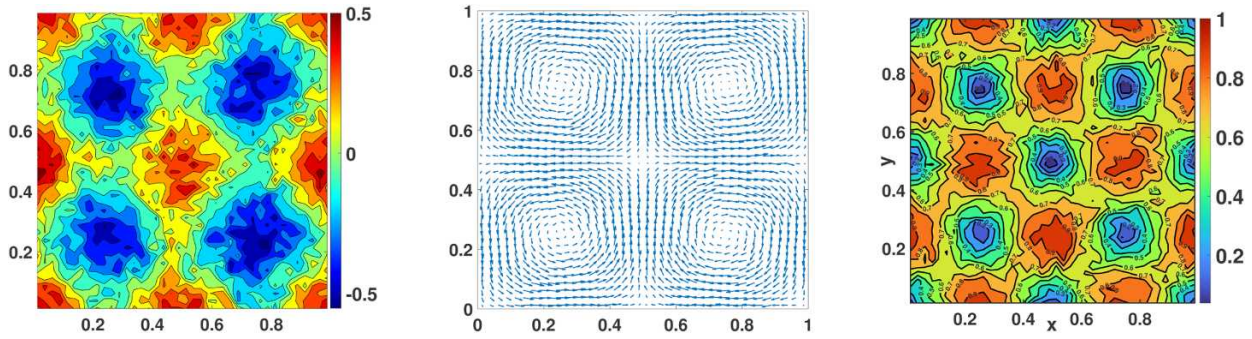


Fig. 18. Estimated state for the Taylor-Green vortex problem with 70% observations and model uncertainty level σ_n^1. (Left) Pressure field. (Center) Velocity field. (Right) Velocity magnitude.

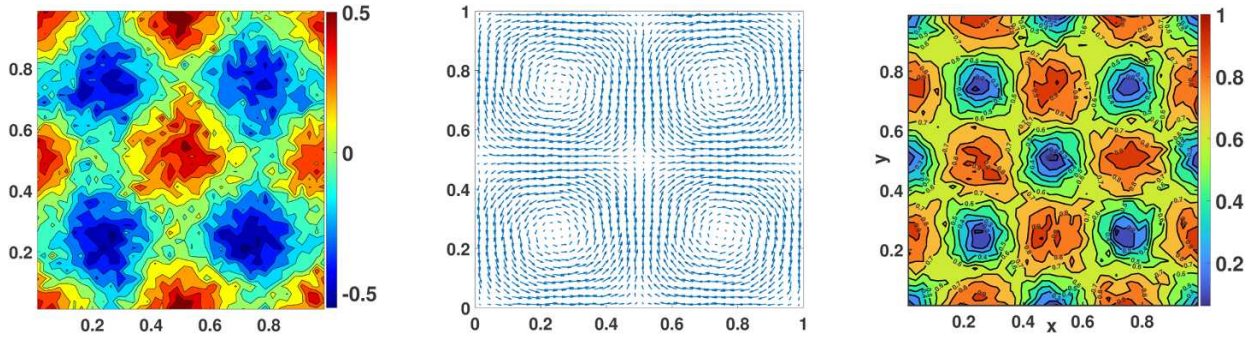


Fig. 19. Estimated state for the Taylor-Green vortex problem with 70% observations and model uncertainty level σ_n^2. (Left) Pressure field. (Center) Velocity field. (Right) Velocity magnitude.

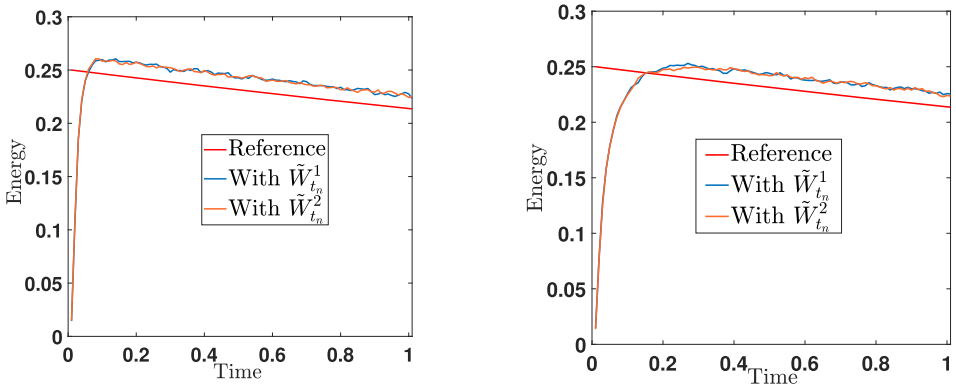


Fig. 20. Estimated energies: (Left) 100% Observations. (Right) 70% Observations.

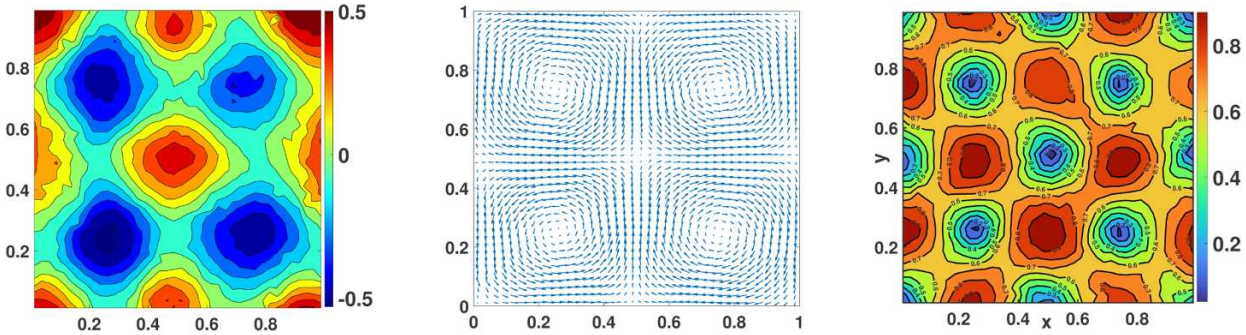


Fig. 21. Estimated state for the Taylor-Green vortex problem by the EnSF with 7% observations and model uncertainty level $\tilde{W}^1_{t_n}$. (Left) Pressure field. (Center) Velocity field. (Right) Velocity magnitude.

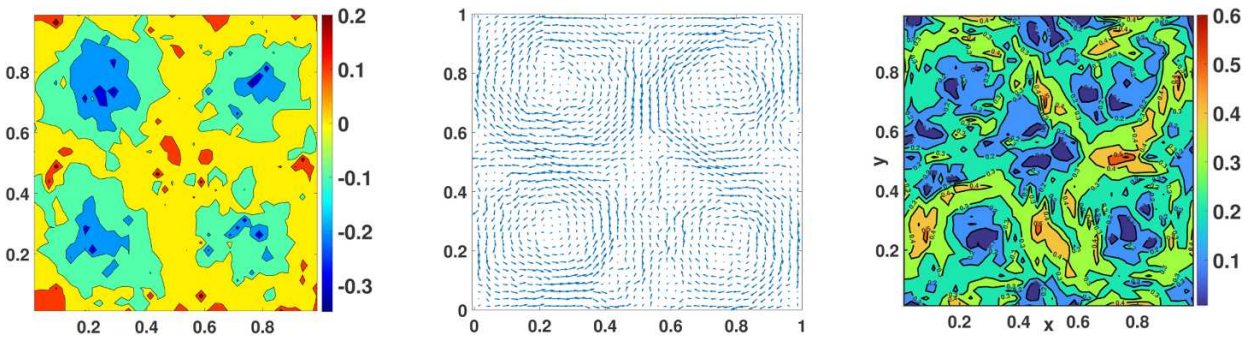


Fig. 22. Estimated state for the Taylor-Green vortex problem with 7% observations by LETKF and model uncertainty level $\tilde{W}^1_{t_n}$. (Left) Pressure field. (Center) Velocity field. (Right) Velocity magnitude.

From the estimated solution figures (e.g., Figs. 21, 22, 24, and 25), we observe that the EnSF consistently produces accurate estimates, regardless of noise level. This accuracy is further confirmed by the RMSE plots in the second panels of Figs. 23 and 26, which show significantly lower errors for EnSF compared to LETKF. Specifically, the pressure and velocity fields estimated by the EnSF closely match the reference solution across most of the domain. In contrast, the LETKF provides only a coarse approximation of the velocity field and fails to recover the pressure field accurately.

Finally, from the energy plots in the first panels of Figs. 23 and 26, it is clear that the LETKF fails to preserve the expected energy dissipation behavior. In contrast, the EnSF estimates closely follow the reference energy curve and exhibit correct dissipative trends.

4.2.2. External force-driven flow

In the second Navier-Stokes test case, we consider a more challenging setting by increasing the Reynolds number to $\mathrm{Re} = 2000$, which results in a more chaotic flow behavior. We consider the same domain $\Omega = (0, 1)^2$ with no-slip boundary conditions on the top and bottom edges and periodic boundary conditions on the two lateral sides. The external body force $\boldsymbol{f}$ is given by $\boldsymbol{f} = \left[f_1, f_2\right]^T$

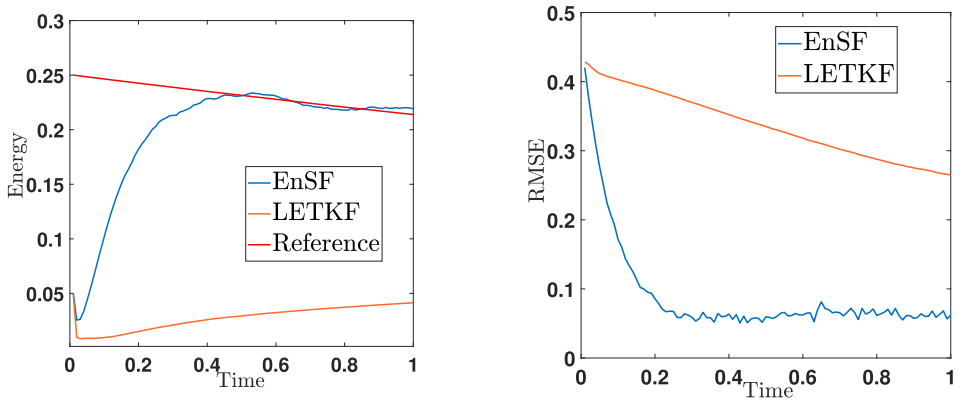


Fig. 23. Estimated energies (Left) and RMSEs for state estimation (Right) with 7% observations and noise $\tilde{W}^1_{t_n}$.

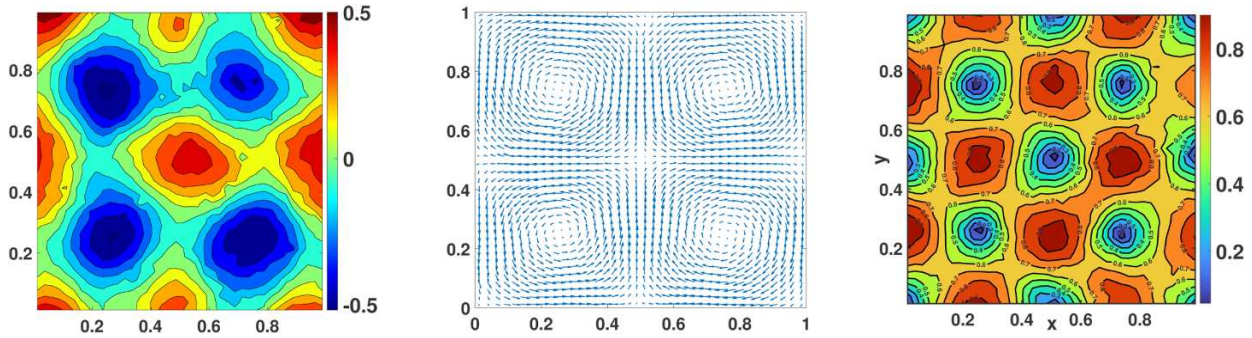


Fig. 24. Estimated state for the Taylor-Green vortex problem by the EnSF with 7% observations and model uncertainty level $\tilde{W}^2_{t_n}$. (Left) Pressure field. (Center) Velocity field. (Right) Velocity magnitude..

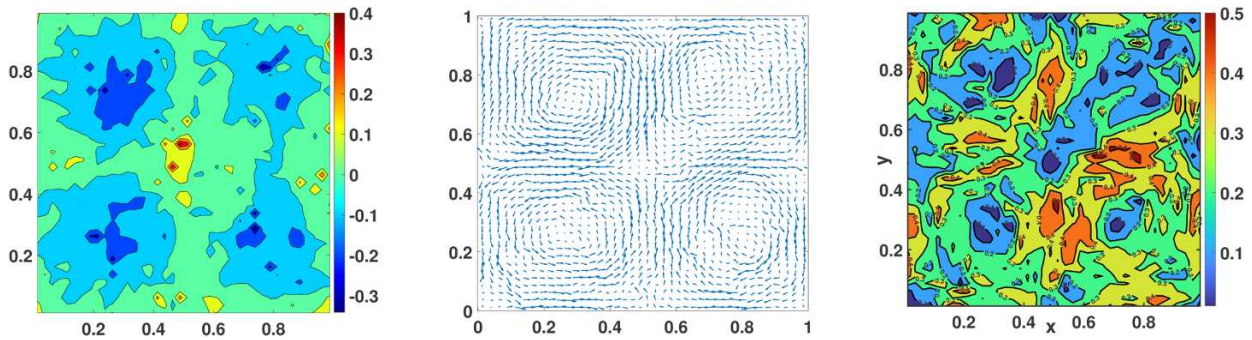


Fig. 25. Estimated state with 7% observations by LETKF and noise $\tilde{W}^2_{t_n}$ (First) Pressure field. (Right) Velocity field. (Third) Velocity magnitude.

where $f_1 = -0.5\cos(8\pi x)$, $f_2 = 0.5\sin(8\pi x)$. Throughout the experiment, we fix the mesh size at $h = 1/80$ and $\theta = 100$. The final time T is $T = 15$, and the time step size for the reference solution is $\Delta t = T/8000$. The reference solutions are shown in Fig. 27.

To carry out the prediction step in data assimilation, we solve the PDE using a coarser time step of $\Delta t_{\text{Filter}} = T/2000$. **The observational data coverage is set to** 20%, and we assume an initial guess for the PDE model given by $\boldsymbol{u}_0 \sim 2 \cdot N(0, \boldsymbol{I}_d)$. At each data assimilation step, the external forcing term $\boldsymbol{f}$ is perturbed by a small noise $\tilde{\omega}$, sampled from $0.0001 \cdot N(0, \boldsymbol{I}_d)$. This test case differs slightly from the previous two, as it accounts not only for the additive term $\sigma_n \tilde{W}_{t_n}$, but also for noise in the external forcing. This introduces a minor modification to the model solver: at each filtering step, the solver applies the stiffness matrix to the perturbed right-hand side rather than the original forcing. The scaling factor σ_n for the model uncertainty $\tilde{W}_{t_n}$ is set to $\sigma_n = 0.005$.

The pressure fields, the velocity fields, and the velocity magnitudes obtained from the EnSF and from the LETKF are shown in Figs. 28 and 29, respectively. These figures demonstrate that our approach recovers the reference solution over nearly the entire region, whereas the LETKF results remain very noisy despite roughly capturing the overall shape. Finally, the RMSEs for both methods

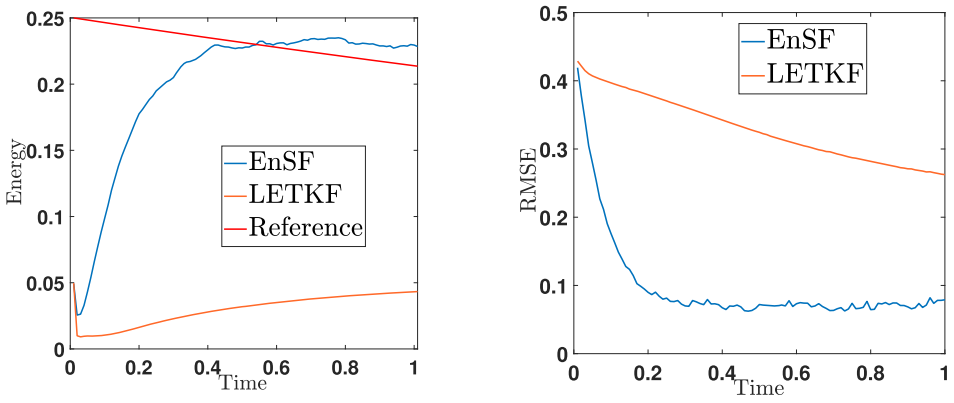


Fig. 26. Estimated energies (Left) and RMSEs for state estimations (Right) with 7% observations and noise $\tilde{W}_{t_n}^2$.

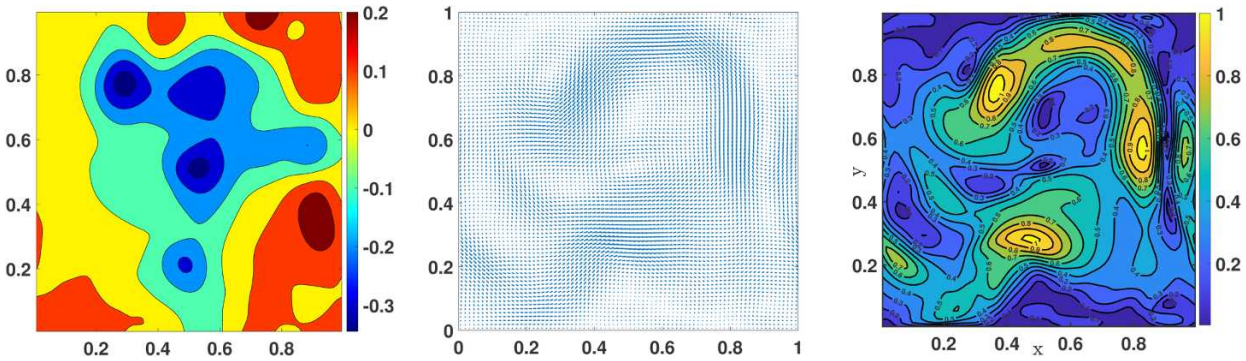


Fig. 27. Reference solution for the NS equation with external force-driven flow. (Left) Pressure field. (Center) Velocity field. (Right) Velocity magnitude.

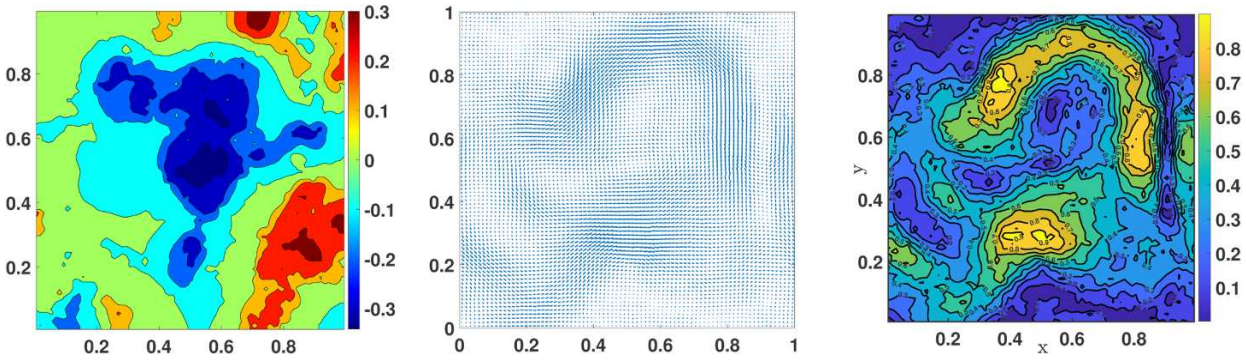


Fig. 28. Estimated state for the force-driven flow problem with 20% observations by EnSF with inpainting. (First) Pressure field. (Right) Velocity field. (Third) Velocity magnitude.

are reported in Fig. 30, which clearly shows that the EnSF yields stable errors over time that remain consistently and substantially lower than those of the LETKF.

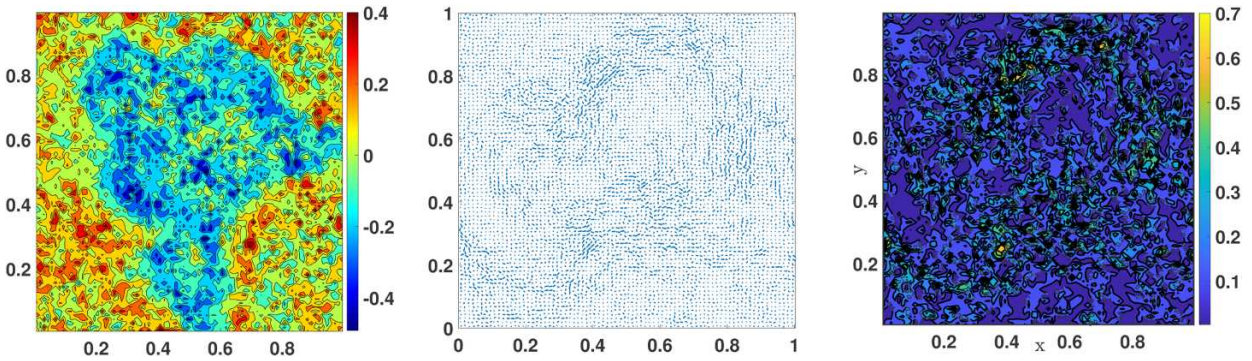


Fig. 29. Estimated state for the force-driven flow problem with 20% observations by LETKF. (First) Pressure field. (Right) Velocity field. (Third) Velocity magnitude.

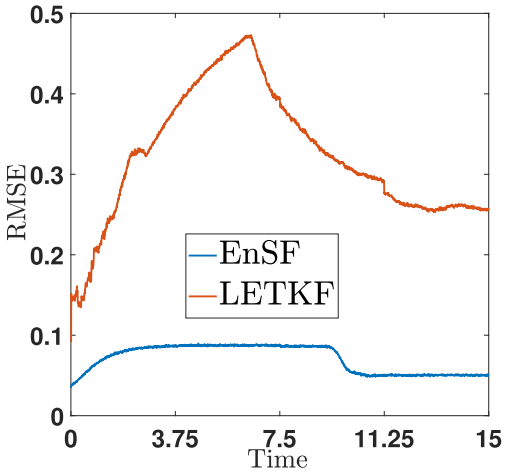


Fig. 30. RMSE for state estimations.

4.3. Allen-Cahn equation

The final example that we use to examine the performance of our adaptive learning procedure is the Allen-Cahn equation with general mobility:

$$\begin{cases} \frac{\partial \phi}{\partial t} = -M(\phi)\mu, & (\boldsymbol{x}, t) \in \Omega \times (0, T], \\ \mu = -\varepsilon^2 \Delta \phi + F'(\phi), & (\boldsymbol{x}, t) \in \Omega \times (0, T], \\ \phi(\boldsymbol{x}, 0) = \phi_0(\boldsymbol{x}), & \boldsymbol{x} \in \Omega \end{cases} \tag{18}$$

and subject to the homogeneous Neumann boundary condition, where $\Omega = (-0.5, 0.5)^2$, $T > 0$ is the terminal time, $\varepsilon > 0$ represents the interfacial width parameter, $M(\phi) \geq 0$ is a general mobility function, and $F(\phi) = \frac{1}{4}\left(1 - \phi^2\right)^2$ is the double-well potential function. For all numerical simulations, we choose $\varepsilon = 0.01$.

We adopt the grain coarsening dynamics test case from [71] with three different scenarios for the mobility:

1. (Case 1) Constant mobility, given by $M(\phi) = M_1(\phi) \equiv 1$.
2. (Case 2) Non-constant but solution-independent mobility, defined as $M(\phi) = M_2(\phi) = \max\{1 + \xi_t, 0\}$, where ξ_t is a perturbed noise.
3. (Case 3) Solution-dependent mobility with noise ξ_t, defined as $M(\phi) = M_3(\phi) = \max\{1 - \phi^2 + \xi_t, 0\}$.

In Case 2 and Case 3, the noise term ξ_t represents the uncertainty in mobility. To generate the synthetic "true solution", we use $\xi_t \sim 0.5 \cdot N(0, \boldsymbol{I}_d)$ in Case 2 and $\xi_t \sim 2 \cdot N(0, \boldsymbol{I}_d)$ in Case 3. However, in the prediction of estimated solutions without observational data – reflecting limited knowledge of the true physical behavior – we intentionally use mis-specified noises: $\xi_t \sim 0.1 \cdot N(0, \boldsymbol{I}_d)$ in Case 2 and $\xi_t \sim 0.4 \cdot N(0, \boldsymbol{I}_d)$ in Case 3.

For this test case, we seek a numerical method that can generate a high-fidelity ground truth while preserving key numerical properties, such as the discrete maximum bound principle (MBP) and unconditional energy stability. Recent methods that achieve these requirements include the family of stabilized integrating factor Runge-Kutta (IFRK) methods [72–74] and an arbitrarily high-order multistep exponential integrator [75]. However, these methods are difficult to extend to problems with variable mobility

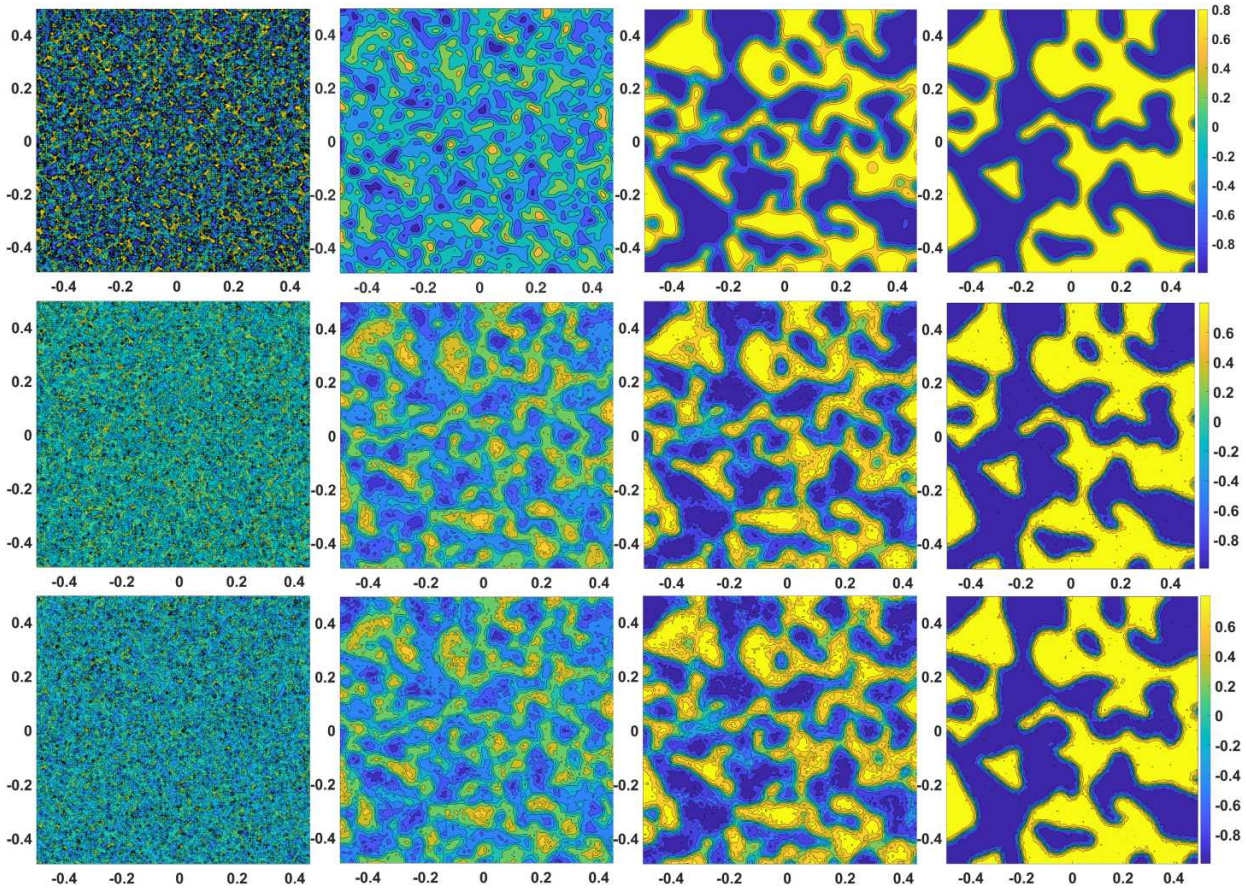


Fig. 31. Solutions of the Allenâ€“Cahn equation in Case 1 at time $t = 0$, $t = \frac{T}{2}$, $t = \frac{2T}{3}$, and $t = T$. (First row) Reference solution. (Second row) Estimated solution with 100% observations. (Third row) Estimated solution with 70% observations.

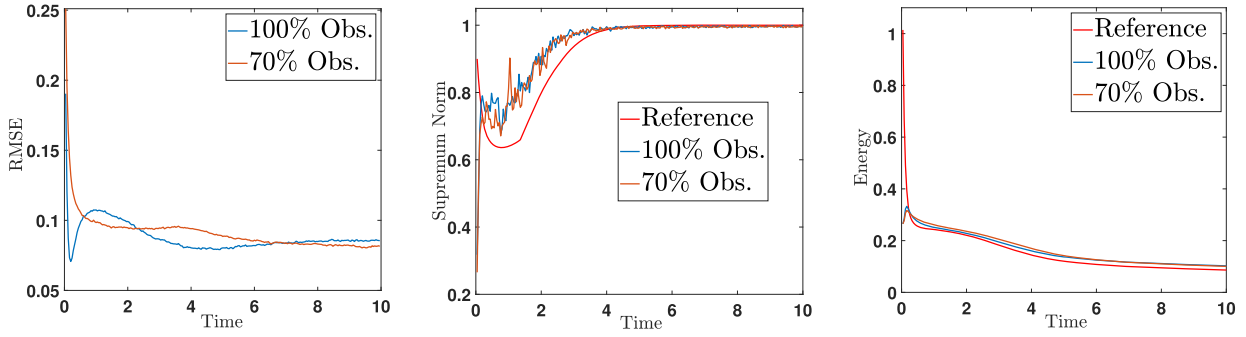


Fig. 32. Illustration of RMSEs, supremum norms, and discrete energies with 100% and 70% observations for Case 1. (Left) RMSE. (Center) Supremum Norm. (Right) Energy.

functions, since their derivations are based on either the variation-of-constants formula or an exponential transformation of the solution. Moreover, solutions of the Allen-Cahn equation (18) typically require a long evolution time to reach steady state and exhibit two distinct phases: an initial rapid transition followed by a much slower convergence to equilibrium [71]. Therefore, it is crucial to employ a high-order, structure-preserving numerical method for the Allen-Cahn equation. To this end, we solve system (18) numerically using the linear, second-order BDF scheme developed in [71]. As shown in [71], this scheme satisfies the discrete MBP and unconditional energy stability, and it is applicable to problems with general mobility functions.

We simulate the experiment on the interval $[0, T]$ with $T = 10$ and discretize the system (18) in space via central finite differences and advance in time with BDF2. This exact solver also serves as the model solver in the prediction step of our data assimilation framework. To reduce computational cost during prediction, one may alternatively use the first-order BDF method proposed in [76].

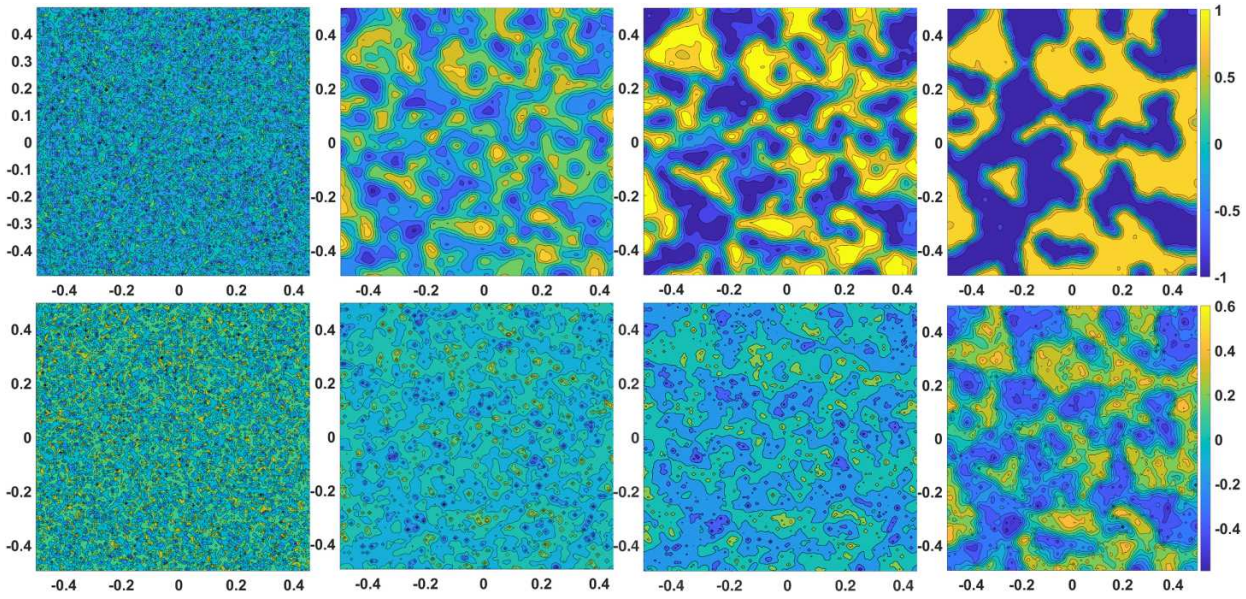


Fig. 33. Solutions of the Allen–Cahn equation in Case 1 at time $t = 0$, $t = \frac{T}{2}$, $t = \frac{2T}{3}$, and $t = T$ with 10% observations. (First row) Estimated solution by EnSF. (Second row) Estimated solution by LETKF.

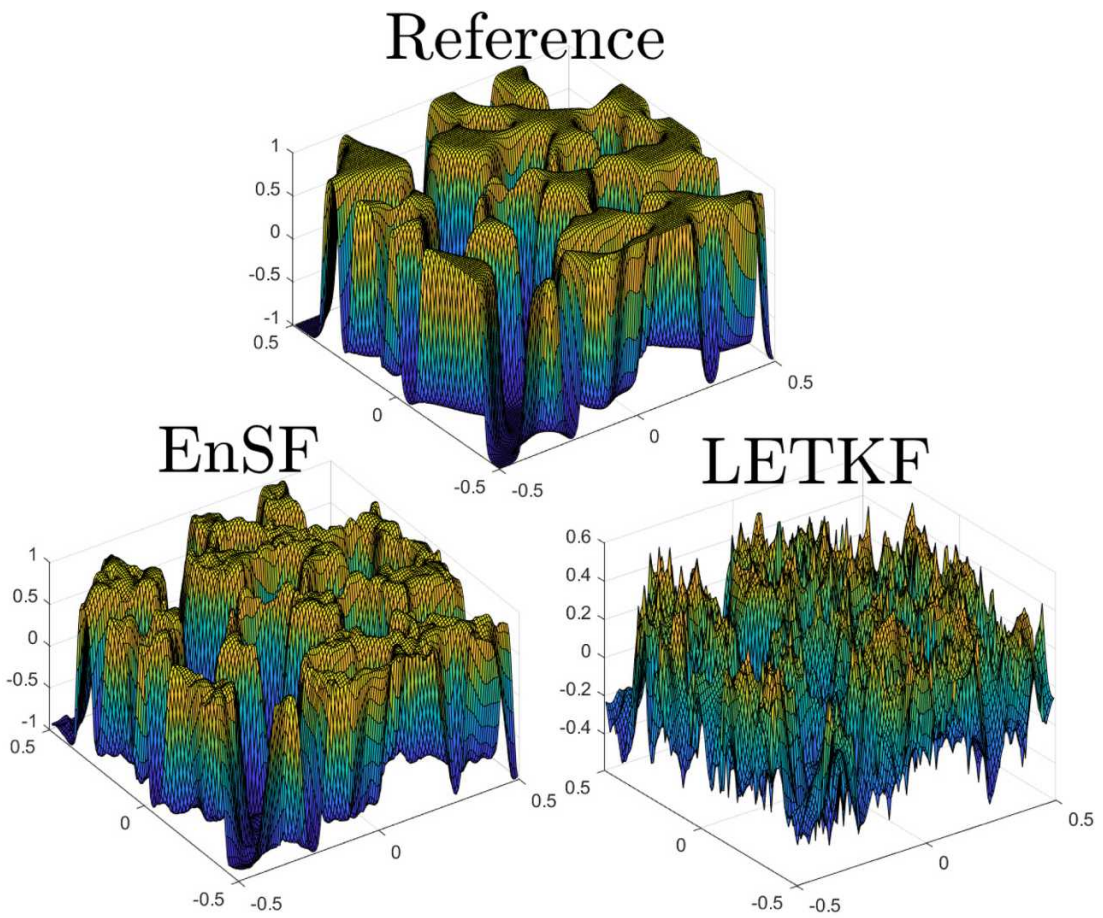


Fig. 34. 3D view of the reference and the estimated solutions for the Allen-Cahn equation in Case 1 at final time T with 10% observations.

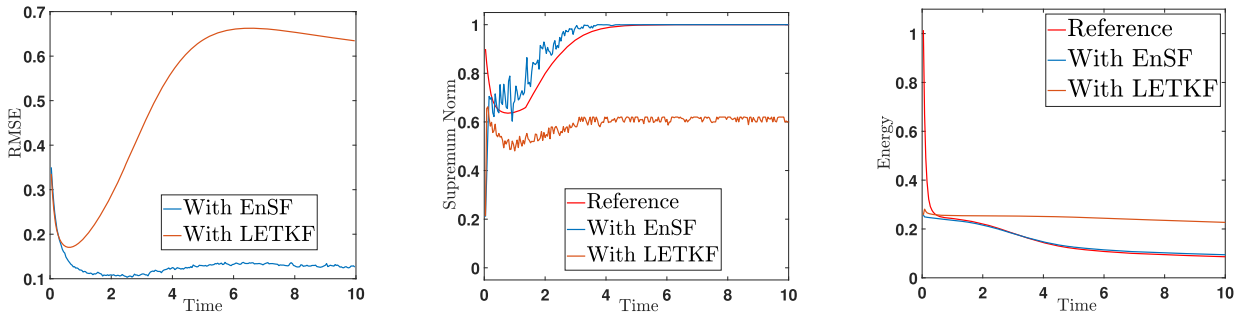


Fig. 35. Illustration of RMSEs, supremum norms, and discrete energies with 10% observations for Case 1. (First) RMSE. (Second) Supremum Norm. (Third) Energy..

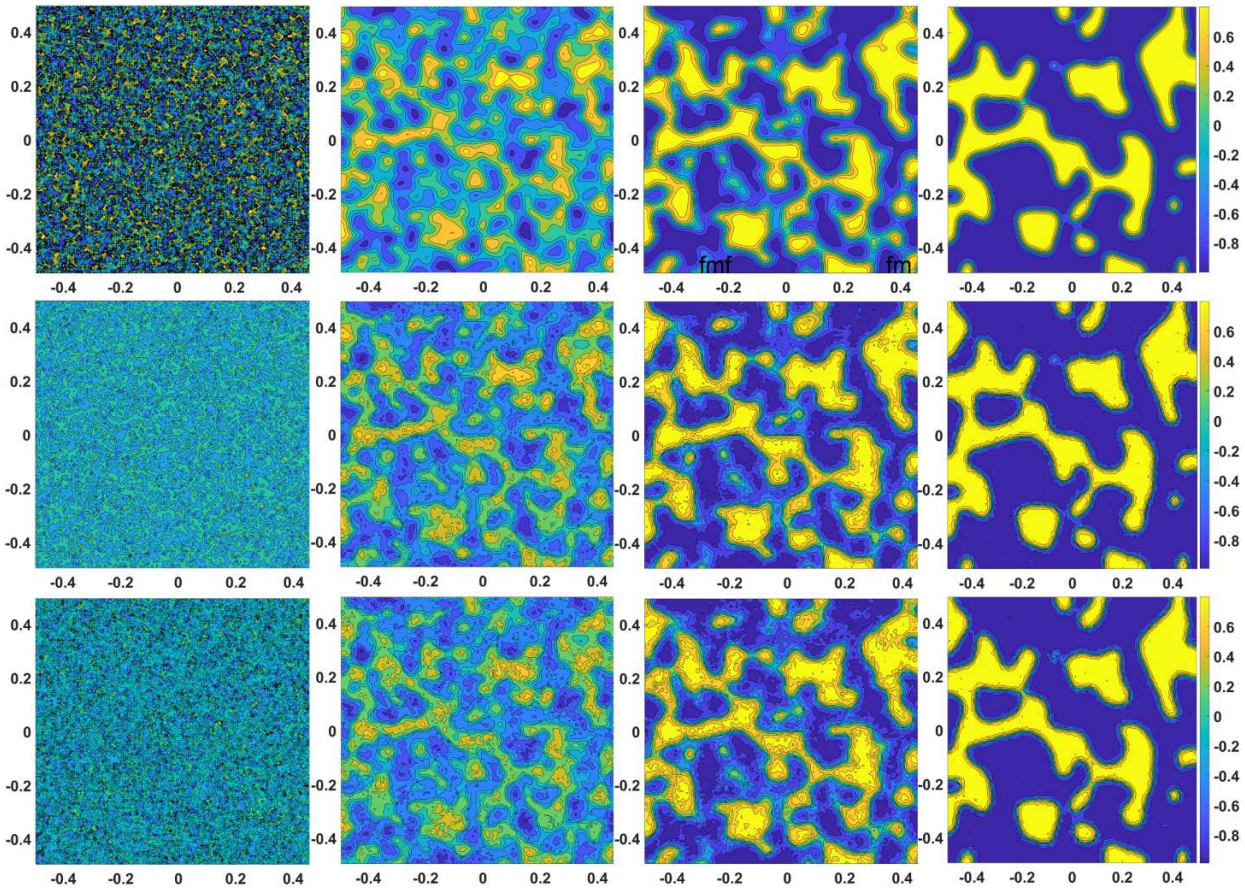


Fig. 36. Solutions of the Allenâ€“Cahn equation in Case 2 at time $t = 0, \frac{T}{2}, \frac{2T}{3}, T$. (First row) Reference. (Second row) Estimated solution with 100% observations. (Third row) Estimated solution with 70% observations.

We note that the algorithm from [71] also provide a robust time-stepping framework that allows non-uniform time-stepping technique. However, this is not the primary focus of our work. Instead, we adopt a fixed and sufficiently small time step $\Delta t = T/1000 = 0.01$ to obtain reliable reference solutions. It has been shown numerically in (18) choosing $\Delta t = 0.01$ uniformly yields comparable results to those obtained with a nonuniform time-stepping strategy. The time step size for the data assimilation process in all cases is fixed at $\Delta t_{\text{Filter}} = T/250$. The spatial mesh size is fixed at $h = 1/128$ for both ground-truth generation and filtering process. Finally, the initial conditions for all cases are generated by randomly sampling the values uniformly from -0.9 to 0.9

For all three test scenarios, **we consider three levels of observational data:** 100%**,** 70%**, and** 10%**. For the** 10% **observation scenario, we additionally compare our approach with the LETKF.** The observational data are received through the ***arctangent operator***. We account for two primary sources of uncertainty: incomplete knowledge of the mobility and model error, the latter represented by the term $\sigma_n \tilde{W}_{t_n}$ in (14). Consequently, because noise is injected into the prediction at each filtering step and different

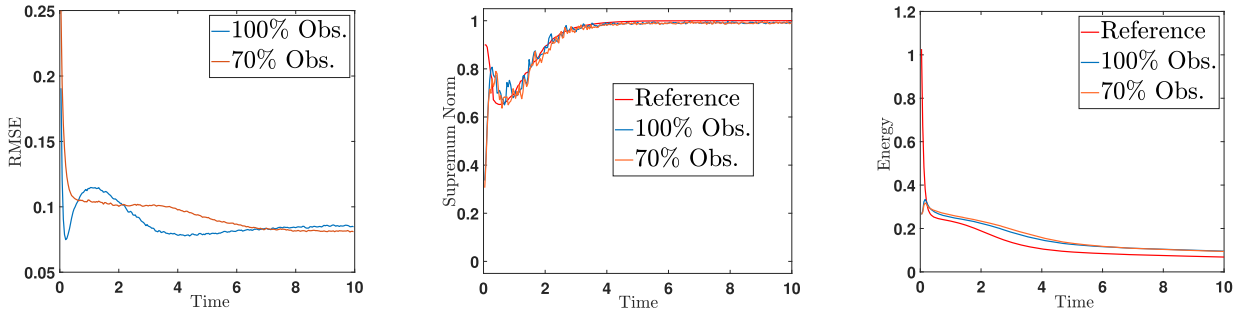


Fig. 37. Illustration of RMSEs, supremum norms, and discrete energies with 100% and 70% observations for Case 2. (Left) RMSE. (Center) Supremum Norm. (Right) Energy..

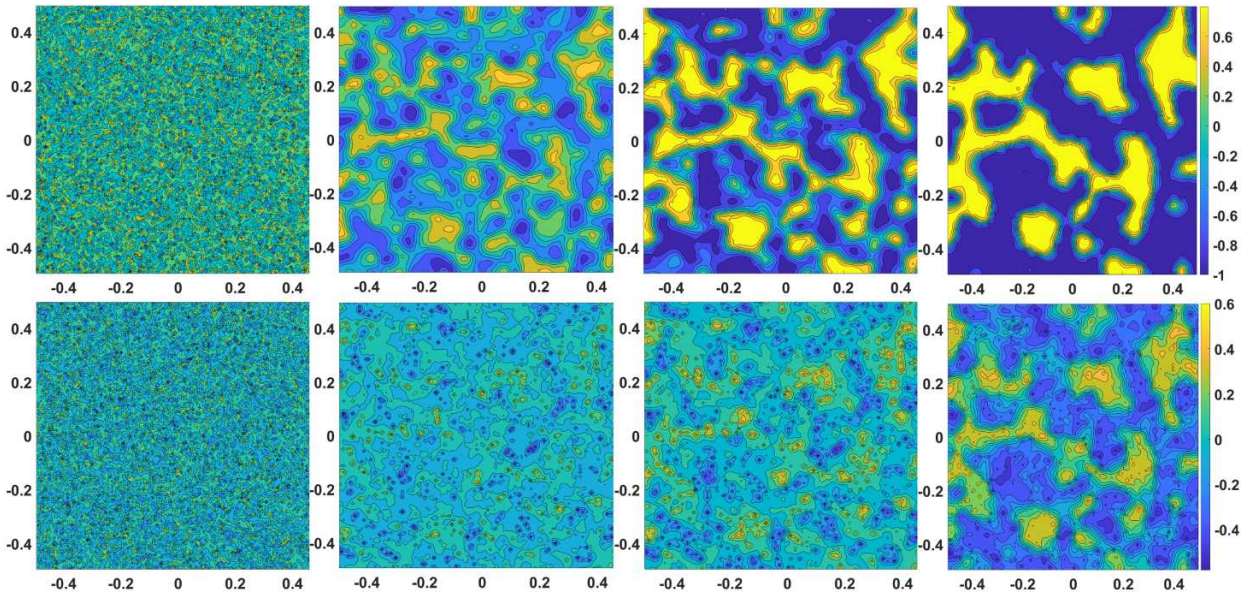


Fig. 38. Solutions of the Allen–Cahn equation in Case 2 at time $t = 0, \frac{T}{2}, \frac{2T}{3}, T$ with 10% observations. (First row) Estimated solution by EnSF. (Second row) Estimated solution by LETKF.

noise intensities are used in computing the stiffness matrix within the data assimilation solver, the resulting solution corresponds to an SPDE whose sample paths generally diverge from those of the reference solution. In all experiments, we fix $\sigma_n = 0.01$, while the level of uncertainty in the mobility varies between scenarios. In addition to the state estimation task, we examine both the discrete energy functional and the supremum norm of the estimates, since it is well-known that the solution to the Allen-Cahn equation satisfies an energy-dissipation law and the maximum-bound principle [71]. Detail of the implementation for the Allen-Cahn equations can be found in https://github.com/Toanhuynh997/StateEst_PDEs/tree/main/AllenCahn.

4.3.1. Case 1: Constant mobility

We first present the results for Case 1. To account for mobility uncertainty, we introduce a small perturbation at each filtering time step, defined as $\eta \sim 0.05 \cdot N(0, \boldsymbol{I}_d)$, which is added to $M_1(\phi)$.

Fig. 31 shows the evolution of the estimated solutions for the 100% and 70% observation scenarios at times $t = 0$, $t = T/2$, $t = 2T/3$, and $t = T$. The corresponding RMSEs, supremum norms, and energy values are presented in Fig. 32.

As shown in Fig. 31, the estimated solutions gradually improve over time and accurately capture the sharp interface of the reference solution, despite starting from different initial conditions. This level of precision is further supported by the RMSE result in the first panel of Fig. 32, which shows a steady decline over time. This demonstrates the effectiveness of our method, particularly given the sensitivity of the Allen-Cahn equation to initial conditions. Finally, the last two panels of Fig. 32 confirm that the estimated solutions adhere to both the maximum bound principle and the energy dissipation property.

We conclude Case 1 by presenting the results for the 10% observation scenario, comparing estimates obtained using EnSF with inpainting and the LETKF. The evolution of the estimated solutions is shown in Fig. 33. It can be observed that our method produces more accurate estimates (top row): the final panel displays a much sharper solution that effectively filters out uncertainty, whereas the LETKF result at time T remains visibly noisy. This observation is further supported by the 3D solution estimation figure depicted

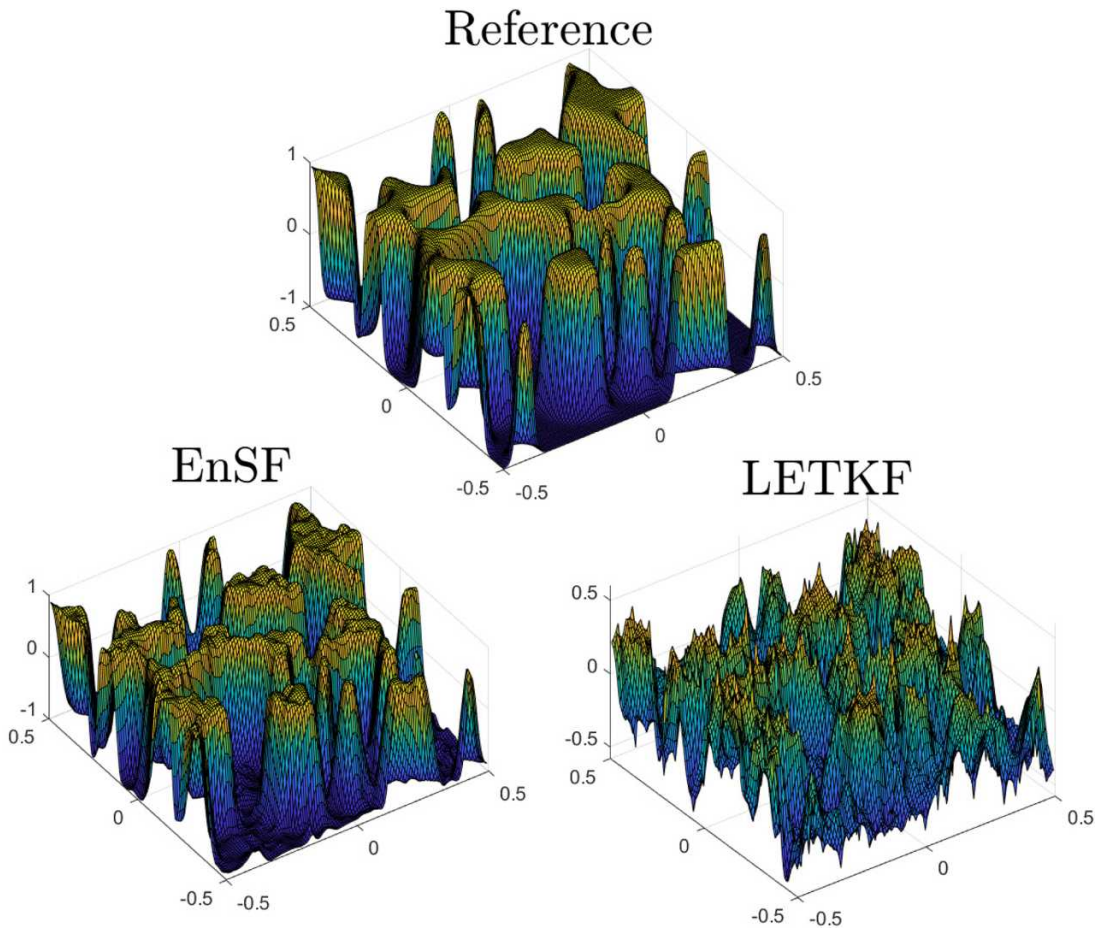


Fig. 39. 3D view of the reference and the estimated solutions for the Allen-Cahn equation in Case 2 at final time T with 10% observations.

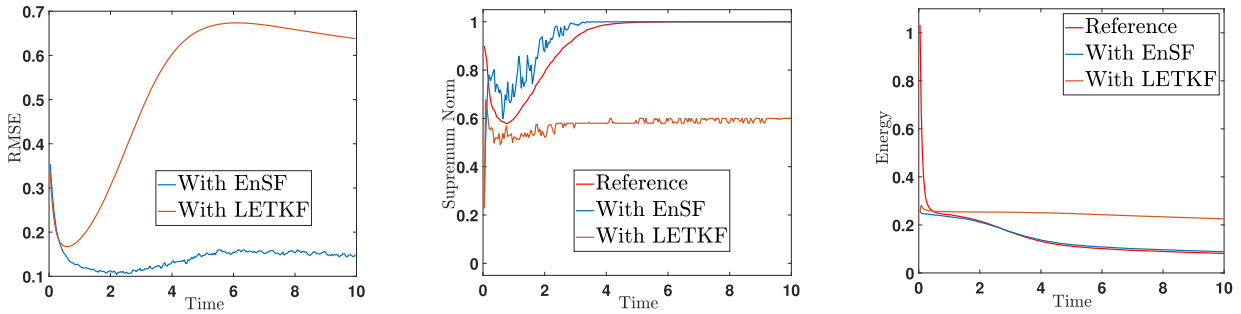


Fig. 40. Illustration of RMSEs, supremum norms, and discrete energies with 10% observations for Case 2. (Left) RMSE. (Center) Supremum Norm. (Right) Energy.

in Fig. 34. Finally, we plot Fig. 35 and compare the estimated solutions in terms of RMSE, supremum norm, and energy. While both methods satisfy the desired theoretical properties, our method outperforms LETKF: it achieves smaller RMSE and its supremum-norm and energy trajectories lie closer to those of the reference solution.

4.3.2. Case 2: Non-constant mobility with uncertainty

To account for the mobility uncertainty in Case 2, we define the mobility as $M_2(\phi) = \max\{1 + \xi_t, 0\}$, where $\xi_t \sim 0.1 \cdot N(0, \boldsymbol{I}_d)$ is the noise used during prediction. The experimental setup for this case follows the same structure as in Case 1.

The evolution of the estimated solutions with 100% and 70% observations are shown in Fig. 36. Despite differences in the initial guess and the presence of various uncertainties in the PDE model, we obtain a consistent state estimation performance in Case 2. In

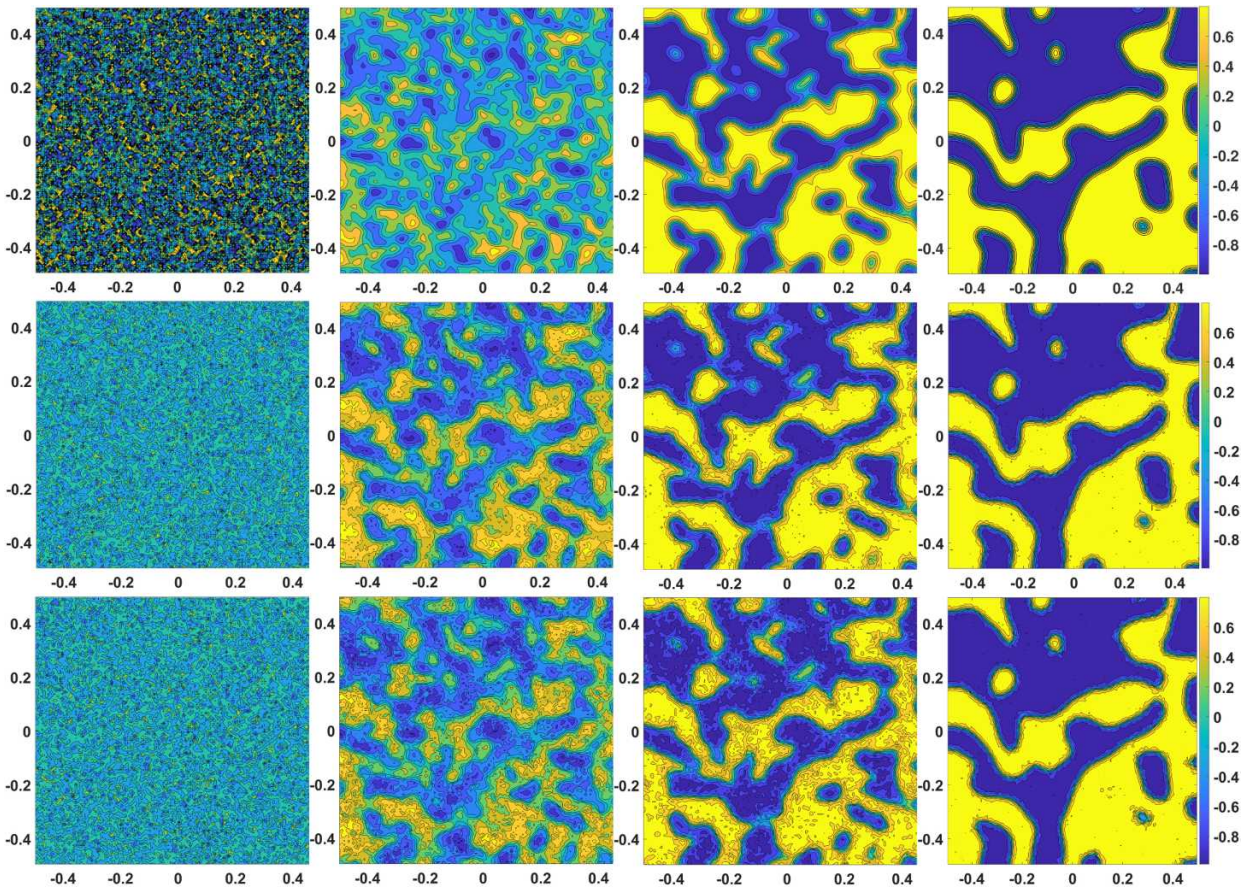


Fig. 41. Solutions of the Allenâ€“Cahn equation in Case 3 at time $t = 0, T/2, 2T/3, T$. (First row) Reference. (Second row) Estimated solution with 100% observations. (Third row) Estimated solution with 70% observations.

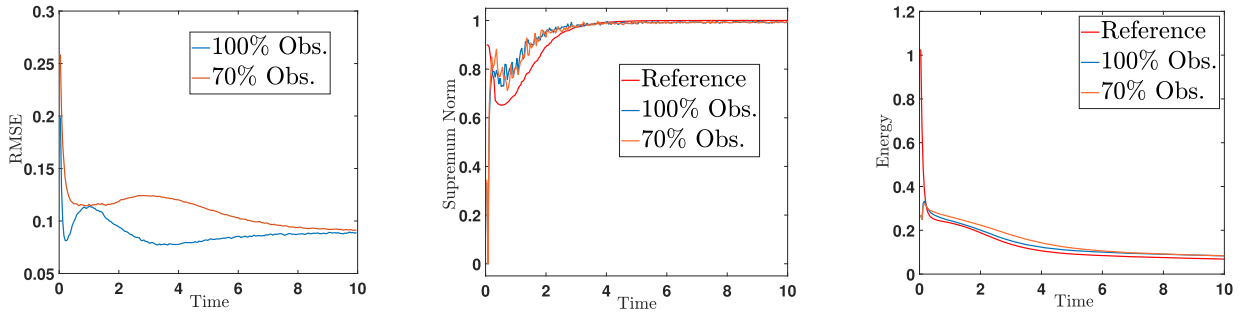


Fig. 42. Illustration of RMSEs, supremum norms, and discrete energies with 100% and 70% observations for Case 3. (Left) RMSE. (Center) Supremum Norm. (Right) Energy.

particular, the estimated solution progressively improves over time and accurately captures the transition interfaces of the reference solution.

The RMSEs, the supremum norms, and the discrete energies of the estimates are displayed in Fig. 37. The first panel of Fig. 37 confirms that the accuracies for both levels of observational data are nearly the same. From the last two panels, it is clear that we still achieve the maximum bound principle and the energy dissipation law for the estimated states.

Finally, we report the results with 10% observations. Estimates produced by the EnSF with inpainting and by the LETKF are shown in Fig. 38, while the 3D views of the estimates at the final time T are plotted in Fig. 39. The corresponding RMSEs, supremum norms, and the discrete energies are depicted in Fig. 40. These figures again confirm that our method outperforms the LETKF in terms of accuracy. Moreover, the estimated solution by our method yields more accurate supremum norm discrete energy.

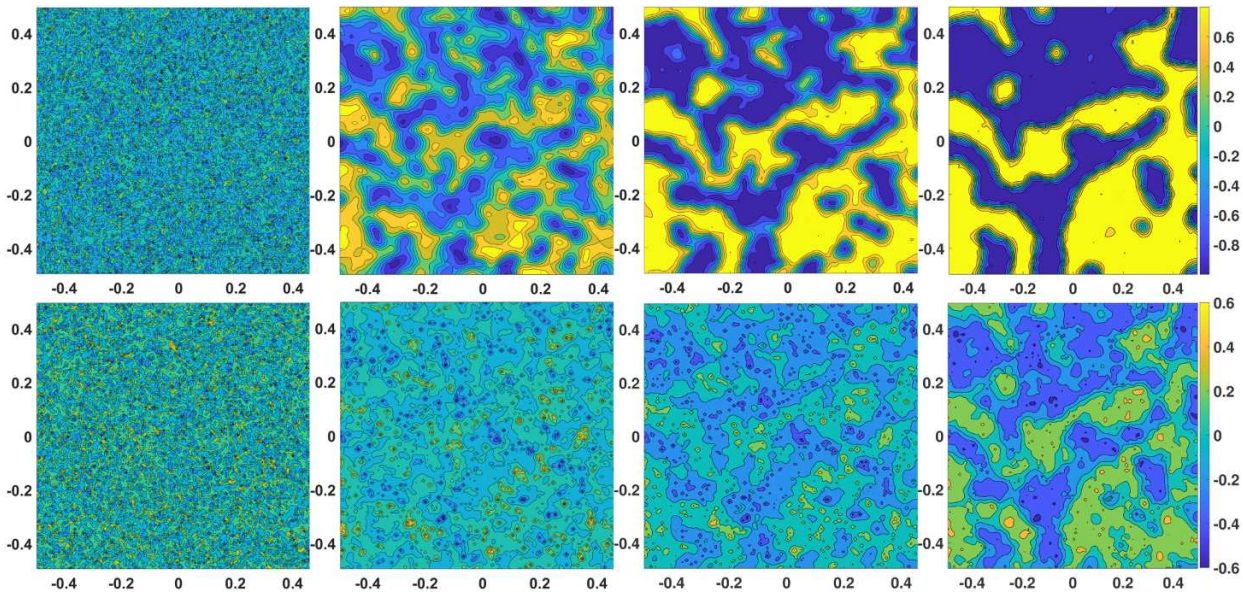


Fig. 43. Solutions of the Allen–Cahn equation in Case 3 at time $t = 0, \frac{T}{2}, \frac{2T}{3}, T$ with 10% observations. (First row) Estimated solution by EnSF. (Second row) Estimated solution by LETKF.

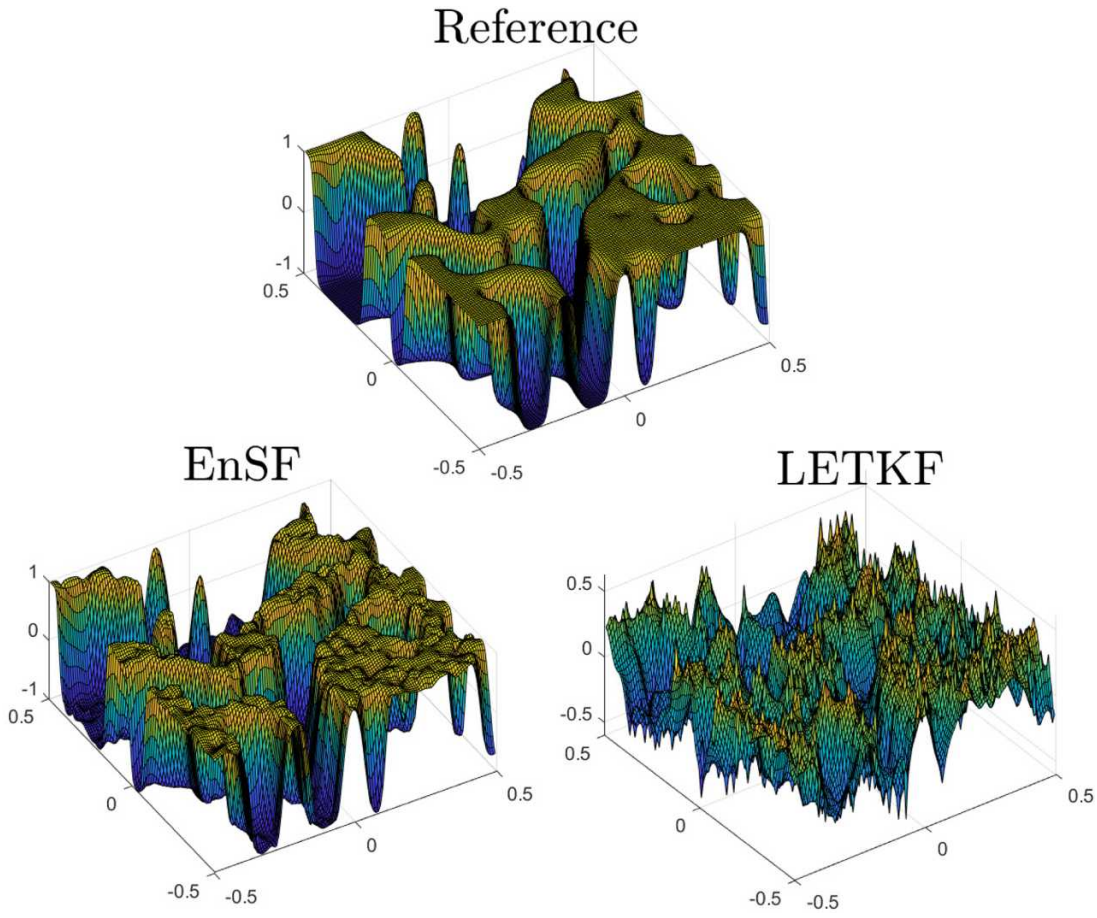


Fig. 44. 3D view of the reference and the estimated solutions for the Allen-Cahn equation in Case 3 at final time T with 10% observatio.

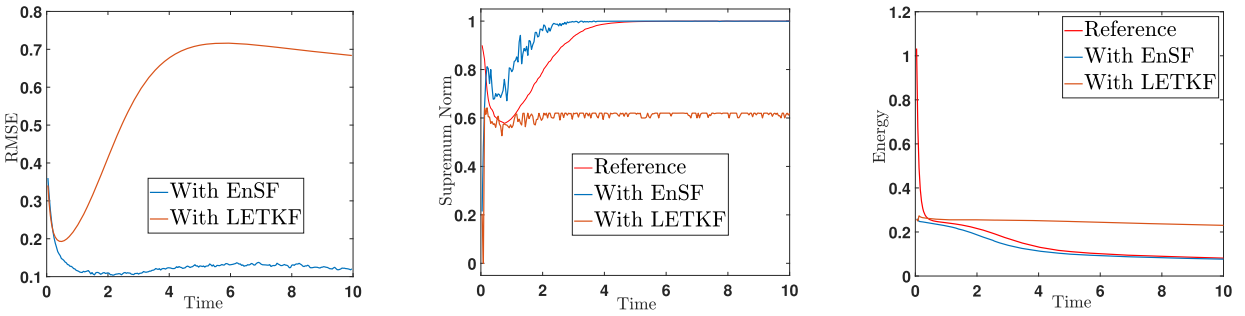


Fig. 45. Illustration of RMSEs, supremum norms, and discrete energies with 10% observations for Case 3. (Left) RMSE. (Center) Supremum Norm. (Right) Energy.

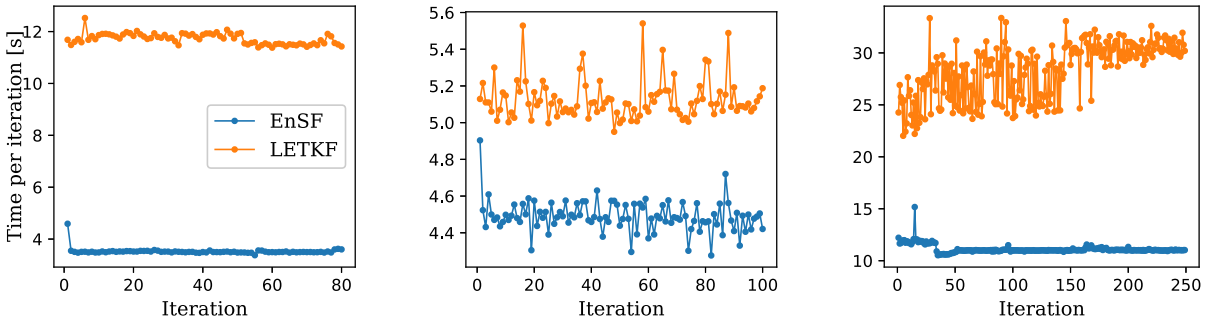


Fig. 46. Illustration of per-iteration wall-clock time for recursive data-assimilation update steps for examples (First) Burguer's equation with terminal time $T = 0.2$, (Second) Taylor green example with noise level $\tilde{W}_{t_n}^2$, and (Third) Allen Cahn with constant mobility. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.3.3. Case 3: Solution-dependent mobility with uncertainty

We conclude the numerical experiments section with Case 3, which involves a solution-dependent mobility subject to noise perturbations. The prediction step in data assimilation is carried out by solving Eq. (18) with the perturbed mobility function $M_3(\phi) = \max\{1 - \phi^2 + \xi_t, 0\}$, where $\xi_t \sim 0.4 \cdot N(0, \boldsymbol{I}_d)$. We evaluate four key quantities: the estimated solution, root-mean-square error (RMSE), supremum norm, and discrete energy.

The results for the 100% and 70% observation scenarios are shown in Figs. 41 and 42, while those for the 10% scenario, including comparisons with the LETKF, are presented in Figs. 43–45. Consistent with the findings in Cases 1 and 2, the results for the solution-dependent mobility with uncertainty case demonstrate that our estimates converge to the reference solution over time, preserve the key properties of the Allenâ€“Cahn equation, and consistently outperform the LETKF in terms of accuracy, supremum norm, and energy.

4.4. Quantitative comparison of computational cost per update step.

Finally, we provide a quantitative comparison of the computational cost between the ensemble score filter (EnSF) and the local ensemble transform Kalman filter (LETKF) by reporting the wall-clock time per recursive Bayesian filtering update step in three test cases. The comparison is restricted to experiments with extremely sparse spatial observations, since these are the cases in which LETKF was also applied. Specifically, we consider three configurations: the 2D Burgers' equation example at terminal time $T = 0.2$ with 10% spatial observations, the 2D Allen-Cahn equation with constant mobility and 10% spatial observations, and the Taylor-Green vortex with noise level $\tilde{W}_{t_n}^2$ and 7% spatial observations. The wall-clock time per filtering time step is plotted in Fig. 46, and the mean of these times is reported in Table 1. Our results demonstrate that the EnSF update step maintains a stable and substantially lower computational cost than LETKF. Combined with the accuracy comparisons presented in the previous sections, these findings indicate that EnSF generally achieves better filtering performance than LETKF at a significantly reduced computational cost, making it a compelling approach for high-dimensional sequential Bayesian framework under extreme data sparsity.

5. Conclusion

In this work, we introduced a novel adaptive learning framework for stochastic partial differential equations (SPDEs) by leveraging score-based diffusion models within a recursive Bayesian inference setting. The proposed methodology encodes the underlying physics

Table 1
Empirical wall-clock time per filtering step for three SPDE experiments. Values report the mean analysis/update time in seconds for the EnSF and LETKF methods, computed over all filtering iterations in each experiment.

Experiment	**EnSF** (s / update step)	**LETKF** (s / update step)
Burgers'	3.52	11.72
Taylor Green	4.49	5.13
Allen-Cahn	11.12	28.32

into the score function using simulation data and incorporates observational information via a likelihood-based correction in a reverse-time SDE. This allows the model to iteratively refine solution estimates as new data becomes available, thereby enhancing predictive accuracy and reducing epistemic uncertainty. Our diffusion model-based approach addresses key limitations of classical optimal filtering methods – such as the dimensionality constraints of particle filters and the linearity assumptions in Kalman-type filters – and enables efficient, physics-informed generative sampling for data assimilation.

We validated the effectiveness and robustness of our approach through comprehensive numerical experiments on benchmark SPDEs, including the Burgersâ$\Delta^{\&}$equations, Navier-Stokes equations, and Allen-Cahn equations under uncertainty. These tests demonstrate the capability of the EnSF to recover hidden physical states and perform accurate solution updates even in the presence of sparse and noisy observations.

Future directions include extending the framework to multi-scale systems and exploring its integration with adaptive mesh refinement strategies for enhanced spatial resolution and computational efficiency.

CRediT authorship contribution statement

Phuoc Toan Huynh: Validation, Data curation; **Ruth Y. López Fajardo:** Data curation; **Guannan Zhang:** Methodology, Investigation, Conceptualization; **Lili Ju:** Methodology, Investigation; **Feng Bao:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Feng Bao reports financial support was provided by US Department of Energy. Feng Bao reports financial support was provided by National Science Foundation. Guannan Zhang reports financial support was provided by US Department of Energy. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This material is based upon work supported by U.S. Department of Energy, Office of Science, Office of Advanced Scientific Computing Research, Applied Mathematics program under the contract ERKJ443 at the Oak Ridge National Laboratory, which is operated by UT-Battelle, LLC, for U.S. Department of Energy under contract DE-AC05-00OR22725. The corresponding author (FB) would like to acknowledge the support from U.S. National Science Foundation through grant DMS-2142672 and from U.S. Department of Energy, Office of Science, Office of Advanced Scientific Computing Research, Applied Mathematics program under grants DE-SC0025412 and DE-SC0024703. L. Ju's research is partially supported by U.S. Department of Energy under grant DE-SC0025527 and U.S. National Science Foundation under grant DMS-2409634. This manuscript has been authored by UT-Battelle, LLC, under contract DE-AC05-00OR22725 with the US Department of Energy (DOE). The US government retains and the publisher, by accepting the article for publication, acknowledges that the US government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this manuscript, or allow others to do so, for US government purposes. DOE will provide public access to these results of federally sponsored research in accordance with the DOE Public Access Plan.

References

[1] R. Abgrall, P.M. Congedo, A semi-intrusive deterministic approach to uncertainty quantification in non-linear fluid flow problems, J. Comput. Phys. 235 (2013) 828–845. https://www.sciencedirect.com/science/article/pii/S0021999112004263. https://doi.org/10.1016/j.jcp.2012.07.041

[2] R. Abgrall, S. Tokareva, The Stochastic Finite Volume Method, Springer International Publishing, Cham, 2017, pp. 1–57. https://doi.org/10.1007/978-3-319-67110-9_1. https://doi.org/10.1007/978-3-319-67110-9_1

[3] T. Barth, On the propagation of statistical model parameter uncertainty in CFD calculations, Theor. Comput. Fluid Dyn. 26 (5) (2012) 435–457.
[4] G. Geraci, P.M. Congedo, R. Abgrall, G. Iaccarino, Intrusive non-linear multiresolution framework for uncertainty quantification in hyperbolic partial differential equations, J. Sci. Comput. 66, (2025) 358–405.
[5] O.M. Knio, O.P. Le Maître, Uncertainty propagation in CFD using polynomial chaos decomposition, Fluid Dyn. Res. 38 (9) (2006) 616–640. Recent Topics in Computational Fluid Dynamics, https://www.sciencedirect.com/science/article/pii/S0169598306000530. https://doi.org/10.1016/j.fluiddyn.2005.12.003
[6] M. Petrella, S. Tokareva, E.F. Toro, Uncertainty quantification methodology for hyperbolic systems with application to blood flow in arteries, J. Comput. Phys. 386 (2019) 405–427. https://www.sciencedirect.com/science/article/pii/S0021999119301305. https://doi.org/10.1016/j.jcp.2019.02.013
[7] S. Walton, S. Tokareva, G. Manzini, The tensor-train stochastic finite volume method for uncertainty quantification, J. Comput. Phys. 538 (2025) 114192. https://www.sciencedirect.com/science/article/pii/S0021999125004759. https://doi.org/10.1016/j.jcp.2025.114192
[8] F. Bao, H.G. Chipilski, S. Liang, G. Zhang, J.S. Whitaker, Nonlinear ensemble filtering with diffusion models: application to the surface quasi-geostrophic dynamics, Mon. Weather Rev. 153 (2025) 1155–1169.
[9] Y. Cheng, A.J. Christlieb, A comparison of stochastic collocation and stochastic Galerkin methods for uncertainty quantification, J. Comput. Phys. 248 (2013) 616–635.
[10] G. Evensen, Data Assimilation: The Ensemble Kalman Filter, Springer Science & Business Media, 2009.
[11] P.T. Huynh, T.-T.-P. Hoang, G. Zhang, F. Bao, Joint state-parameter estimation for the reduced fracture model via the united filter, J. Comput. Phys. 538 (2025) 114159.
[12] T. Huynh, F. Bao, T. Hoang, Parameter estimation for the reduced fracture model via a direct filter method, J. Mach. Learn. Model. Comput. 6 (1) (2025) 23–40.
[13] M.C. Kennedy, A. O'Hagan, Bayesian calibration of computer models, J. R. Stat. Soc. Ser. B (Stat. Methodol.) 63 (3) (2001) 425–464.
[14] D.S. Oliver, A.C. Reynolds, N. Liu, Inverse Theory for Petroleum Reservoir Characterization and History Matching, Cambridge University Press, 2008.
[15] S. Reich, C. Cotter, Probabilistic Forecasting and Bayesian Data Assimilation, Cambridge University Press, 2015.
[16] M. Buehner, Local ensemble transform Kalman filter with cross validation, MWR 148 (2020) 2265–2282. https://doi.org/10.1175/MWR-D-19-0402.1
[17] R.E. Kalman, A new approach to linear filtering and prediction problems. J. Basic Eng. 82 (1) (1960) 35–45 https://doi.org/10.1115/1.3662552.
[18] C. Andrieu, A. Doucet, R. Holenstein, Particle Markov chain Monte Carlo methods, J. R. Stat. Soc. B 72 (3) (2010) 269–342.
[19] R. Archibald, F. Bao, X. Tu, A direct method for parameter estimations, J. Comput. Phys. 398 (2019) 108871.
[20] N.J. Gordon, D.J. Salmond, A.F.M. Smith, Novel approach to nonlinear/non-Gaussian Bayesian state estimation, IEE Proc.-F 140 (2) (1993) 107–113.
[21] K. Kang, V. Maroulas, I. Schizas, F. Bao, Improved distributed particle filters for tracking in a wireless sensor network, Comput. Stat. Data Anal. 117 (2018) 90–108.
[22] M. Morzfeld, X. Tu, E. Atkins, A.J. Chorin, A random map implementation of implicit filters, J. Comput. Phys. 231 (4) (2012) 2049–2066.
[23] M.K. Pitt, N. Shephard, Filtering via simulation: auxiliary particle filters, J. Am. Stat. Assoc. 94 (446) (1999) 590–599.
[24] R. Archibald, F. Bao, Kernel learning backward SDE filter for data assimilation, J. Comput. Phys. 455 (3) (2022) 111009. https://doi.org/10.1016/j.jcp.2022.111009
[25] F. Bao, Y. Cao, X. Han, An implicit algorithm of solving nonlinear differential equations and optimal filtering of diffusion processes, Commun. Math. Sci. 18 (3) (2014) 635–661. https://doi.org/10.4310/CMS.2020.v18.n3.a3
[26] F. Bao, Y. Cao, C. Webster, G. Zhang, A hybrid sparse-grid approach for nonlinear filtering problems based on adaptive-domain of the Zakai equation approximations, SIAM/ASA J. Uncertain. Quantif. 2 (1) (2014) 784–804. https://doi.org/10.1137/140952910. https://doi.org/10.1137/140952910
[27] F. Bao, Y. Cao, W. Zhao, Numerical solutions for forward backward doubly stochastic differential equations and Zakai equations, Visualization Mech. Processes: Int. Online J. 1 (4) (2011) 351–367.
[28] M. Zakai, On the optimal filtering of diffusion processes, Z. Wahrscheinlichkeitstheorie und Verw. Gebiete 11 (1969) 230–243.
[29] Z. Zhang, R. Archibald, F. Bao, A PDE-based adaptive kernel method for solving optimal filtering problems, J. Mach. Learn. Model. Comput. 3 (2022) 37–59.
[30] J. Ho, A. Jain, P. Abbeel, Denoising diffusion probabilistic models, Adv. Neural Inf. Process. Syst. 33 (2020) 6840–6851.
[31] S. Liang, H. Tran, F. Bao, H.G. Chipilski, P.J. van Leeuwen, G. Zhang, Ensemble score filter with image inpainting for data assimilation in tracking surface quasi-geostrophic dynamics with partial observations, arXiv:2501.12419.
[32] Y. Song, J. Sohl-Dickstein, D.P. Kingma, A. Kumar, S. Ermon, B. Poole, Score-based generative modeling through stochastic differential equations, in: International Conference on Learning Representations, 2021. https://openreview.net/forum?id = PxTIG12RRHS.
[33] B.R. Hunt, E.J. Kostelich, I. Szunyogh, Efficient data assimilation for spatiotemporal chaos: a local ensemble transform Kalman filter, Phys. D 230 (1-2) (2007) 112–126.
[34] C.-C. Hu, P.J. van Leeuwen, A particle flow filter for high-dimensional system applications, QJRMS 147 (2021) 2352-2374. https://doi.org/10.1002/qj.4028
[35] C. Snyder, T. Bengtsson, P. Bickel, J. Anderson, Obstacles to high-dimensional particle filtering, Mon. Wea. Rev. 136 (2008) 4629–4640.
[36] Y. Song, S. Ermon, Generative modeling by estimating gradients of the data distribution, in: Proc. NIPS 2019, Vancouver, Canada, 2019, pp. 11918–11930.
[37] F. Bao, Z. Zhang, G. Zhang, An ensemble score filter for tracking high-dimensional nonlinear dynamical systems, Comput. Methods Appl. Mech. Eng. 432 (2024) 117447. https://www.sciencedirect.com/science/article/pii/S0045782524007023. https://doi.org/10.1016/j.cma.2024.117447
[38] F. Bao, Z. Zhang, G. Zhang, A score-based filter for nonlinear data assimilation, J. Comput. Phys. 514 (2024) 113207. https://www.sciencedirect.com/science/article/pii/S002199912400456X. https://doi.org/10.1016/j.jcp.2024.113207
[39] D.A. Barajas-Solano, D.M. Tartakovsky, Stochastic collocation methods for nonlinear parabolic equations with random coefficients, SIAM/ASA J. Uncertainty Quantif. 4 (1) (2016) 475–494. https://doi.org/10.1137/130930108. https://doi.org/10.1137/130930108
[40] J.J. Harmon, S. Tokareva, A. Zlotnik, P.J. Swart, Adaptive uncertainty quantification for stochastic hyperbolic conservation laws, SIAM/ASA J. Uncertainty Quantif. 13 (2) (2025) 339–374. https://doi.org/10.1137/23M1624750. https://doi.org/10.1137/23M1624750
[41] J. Jung, H. Shin, M. Choi, Bayesian deep learning framework for uncertainty quantification in stochastic partial differential equations, SIAM J. Sci. Comput. 46 (1) (2024) C57–C76. https://doi.org/10.1137/23M1560574. https://doi.org/10.1137/23M1560574
[42] S. Mo, N. Zabaras, X. Shi, J. Wu, Deep autoregressive neural networks for highdimensional inverse problems in groundwater contaminant source identification, Water Resour. Res. 55 (2019) 3856–3881.
[43] C. S, M. Choi, MGDGAN: multiple generator and discriminator generative adversarial networks for solving stochastic partial differential equations, IEEE Access 10 (2022) 130908–130920. https://doi.org/10.1109/ACCESS.2022.3229696
[44] S. Tokareva, A. Zlotnik, V. Gyrya, Stochastic finite volume method for uncertainty quantification of transient flow in gas pipeline networks, Appl. Math. Model. 125 (2024) 66–84. https://www.sciencedirect.com/science/article/pii/S0307904X23004249. https://doi.org/10.1016/j.apm.2023.09.017
[45] Q. Zhao, Y. Zhao, J. Bai, J. Yan, J. Wang, S.V. Ablameyko, Uncertainty quantification of stochastic partial differential equations based on bayesian neural networks, in: 2024 International Conference on New Trends in Computational Intelligence (NTCI), 2024, pp. 320–325. https://doi.org/10.1109/NTCI64025.2024.10776066
[46] A.H.M. Rafid, J. Yin, Y. Geng, S. Liang, F. Bao, L. Ju, G. Zhang, A scalable training-free diffusion model for uncertainty quantification, in: Proceedings of the SC '24 Workshops of the International Conference on High Performance Computing, Network, Storage, and Analysis, 2024.
[47] J. Yin, S. Liang, S. Liu, F. Bao, H.G. Chipilski, D. Lu, G. Zhang, A scalable real-time data assimilation framework for predicting turbulent atmosphere dynamics, in: Proceedings of the SC '24 Workshops of the International Conference on High Performance Computing, Network, Storage, and Analysis, SC-W '24, IEEE Press, 2024, p. 11-18.
[48] C.-W. Shu, S. Osher, Efficient implementation of essentially non-oscillatory shock-capturing schemes, J. Comput. Phys. 77 (2) (1988) 439–471.
[49] B. Cockburn, C.W. Shu, The Runge-Kutta local projection-discontinuous-Galerkin finite element method for scalar conservation laws, ESAIM: Math. Model. Numer. Anal. 25 (3) (1991) 337–361.
[50] S. Gottlieb, C.W. Shu, Total variation diminishing Runge-Kutta schemes, Math. Comput. 67 (221) (1998) 73–85.
[51] B. Cockburn, Devising discontinuous Galerkin methods for non-linear hyperbolic conservation laws, J. Comput. Appl. Math. 128 (1-2) (2001) 187–204.

[52] C.-K. Doan, P.T. Huynh, T.T.P. Hoang, Noniterative localized exponential time differencing methods for hyperbolic conservation laws, Adv. Comput. Mat. 51 (26) (2025). https://doi.org/10.1007/s10444-025-10240-0
[53] A. Harten, B. Engquist, S. Osher, S.R. Chakravarthy, Uniformly high order accurate essentially non-oscillatory schemes III, J. Comput. Phys. 71 (1987) 231–303.
[54] F. Hu, The 6th-order weighted ENO schemes for hyperbolic conservation laws, Comput. Fluids 174 (2018) 34–45.
[55] G.S. Jiang, C.W. Shu, Efficient implementation of weighted ENO schemes, J. Comput. Phys. 126 (1) (1996) 202–228.
[56] Y. Jiang, C.-W. Shu, M. Zhang, An alternative formulation of finite difference weighted ENO schemes with Lax-Wendroff time discretization for conservation laws, SIAM J. Sci. Comput. 35 (2) (2013) A1137–A1160.
[57] A.J. Kozakevicius, L.C.C. Santos, ENO adaptive method for solving one-dimensional conservation laws, Appl. Numer. Math. 59 (9) (2009) 2337-2355.
[58] X.D. Liu, S. Osher, T. Chan, Weighted essentially non-oscillatory schemes, J. Comput. Phys. 115 (1) (1994) 200–212.
[59] C.-W. Shu, S. Osher, Efficient implementation of essentially non-oscillatory shock-capturing schemes II, J. Comput. Phys. 83 (1) (1989) 32-78.
[60] G. Akrivis, B. Li, D. Li, Energy-decaying extrapolated RK-SAV methods for the Allen-Cahn and Cahn-Hilliard equations, SIAM J. Sci. Comput. 41 (6) (2019) A3703–A3727.
[61] Q. Cheng, C. Wang, Error estimate of a second order accurate scalar auxiliary variable (SAV) numerical method for the epitaxial thin film equation, Adv. Appl. Math. Mech. 13 (6) (2021) 1318–1354.
[62] M. Jiang, Z. Zhang, J. Zhao, Improving the accuracy and consistency of the scalar auxiliary variable (SAV) method with relaxation, J. Comput. Phys. 456 (2022) 110954.
[63] J. Shen, J. Xu, J. Yang, The scalar auxiliary variable (SAV) approach for gradient flows, J. Comput. Phys. 353 (2018) 407–416.
[64] J. Shen, J. Xu, J. Yang, A new class of efficient and robust energy stable schemes for gradient flows, SIAM Rev. 61 (3) (2019) 474–506.
[65] M. Wang, Q. Huang, C. Wang, A second order accurate scalar auxiliary variable (SAV) numerical method for the square phase field crystal equation, J. Sci. Comput. 88 (2) (2021) 33.
[66] Q. Cheng, C. Liu, J. Shen, A new Lagrange multiplier approach for gradient flows, Comput. Methods Appl. Mech. Eng. 367 (2020) 113070.
[67] J. Yang, Z. Tan, J. Kim, Original variables based energy-stable time-dependent auxiliary variable method for the incompressible Navier-Stokes equation, Comput. Fluids 240 (2022) 105432.
[68] C.-K. Doan, T.T.P. Hoang, L. Ju, R. Lan, Dynamically regularized Lagrange multiplier schemes with energy dissipation for the incompressible Navier-Stokes equations, J. Comput. Phys. 521 (2025) 113550. https://doi.org/10.1016/j.jcp.2024.113550
[69] X. Li, J. Shen, Error analysis of the SAV-MAC scheme for the NS equations, SIAM J. Numer. Anal. 58 (5) (2020) 2465–2491.
[70] J.L. Guermond, P. Minev, J. Shen, An overview of projection methods for incompressible flows, Comput. Methods Appl. Mech. Eng. 195 (44–47) (2006) 6011–6045.
[71] D. Hou, L. Ju, Z. Qiao, A linear second-order maximum bound principle-preserving BDF scheme for the Allen-Cahn equation with a general mobility, Math. Comput. 92 (344) (2023) 2515–2542.
[72] J. Li, X. Li, L. Ju, X. Feng, Stabilized integrating factor Runge-Kutta method and unconditional preservation of maximum bound principle, SIAM J. Sci. Comput. 43 (3) (2021) A1780–A18021.
[73] H. Zhang, J. Yan, X. Qian, X. Gu, S. Song, On the preserving of the maximum principle and energy stability of high-order implicit-explicit Runge-Kutta schemes for the space-fractional Allen-Cahn equation, Numer. Algorithms 88 (3) (2021) 1309–1336.
[74] H. Zhang, J. Yan, X. Qian, S. Song, Numerical analysis and applications of explicit high order maximum principle preserving integrating factor Runge-Kutta schemes for AllenCahn equation, Appl. Numer. Math. 161 (2021) 372–390.
[75] B. Li, J. Yang, Z. Zhou, Arbitrarily high-order exponential cut-off methods for preserving maximum principle of parabolic equations, SIAM J. Sci. Comput. 42 (6) (2020) A3957–3978.
[76] J. Shen, T. Tang, J. Yang, On the maximum principle preserving schemes for the generalized Allen-Cahn equation, Commun. Math. Sci. 14 (6) (2016) 1517–1534.